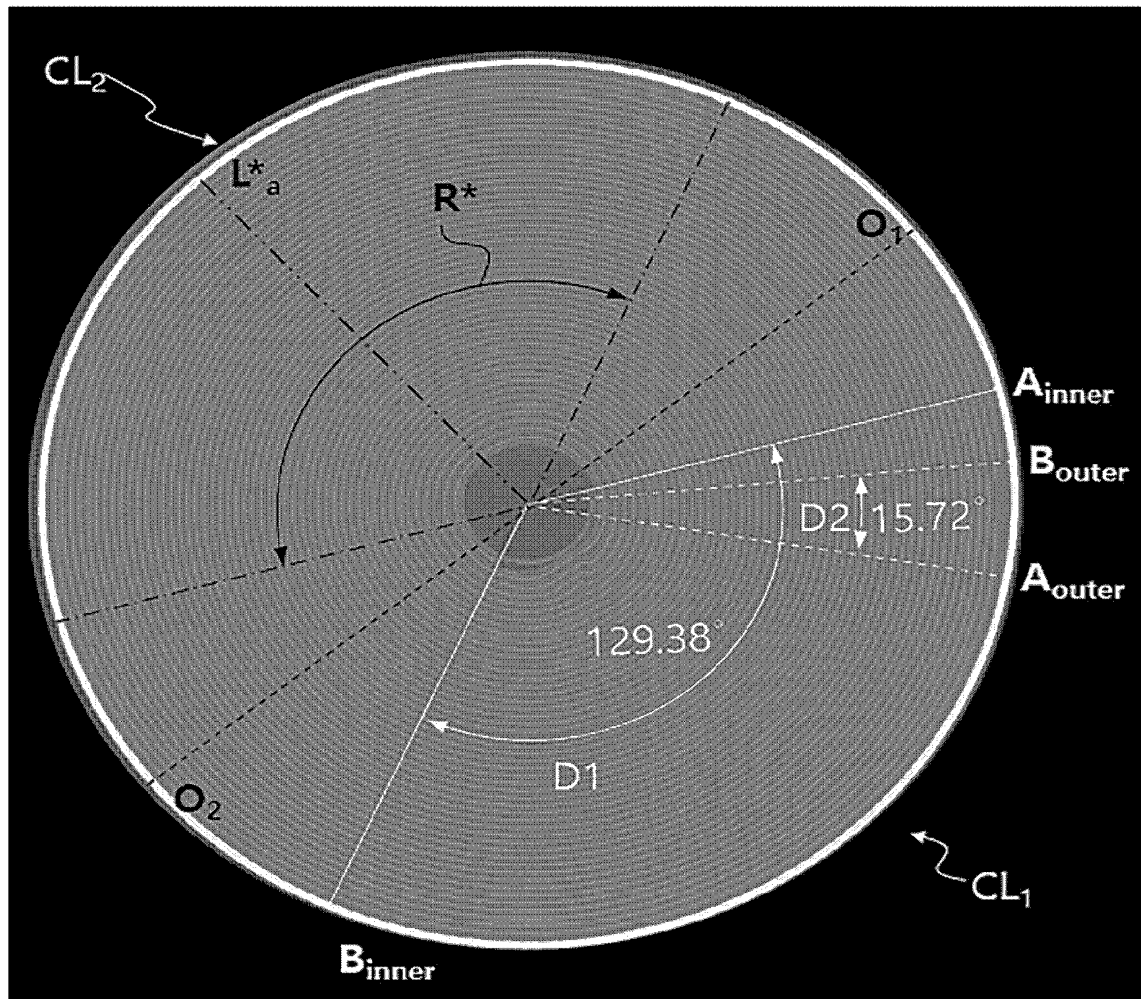
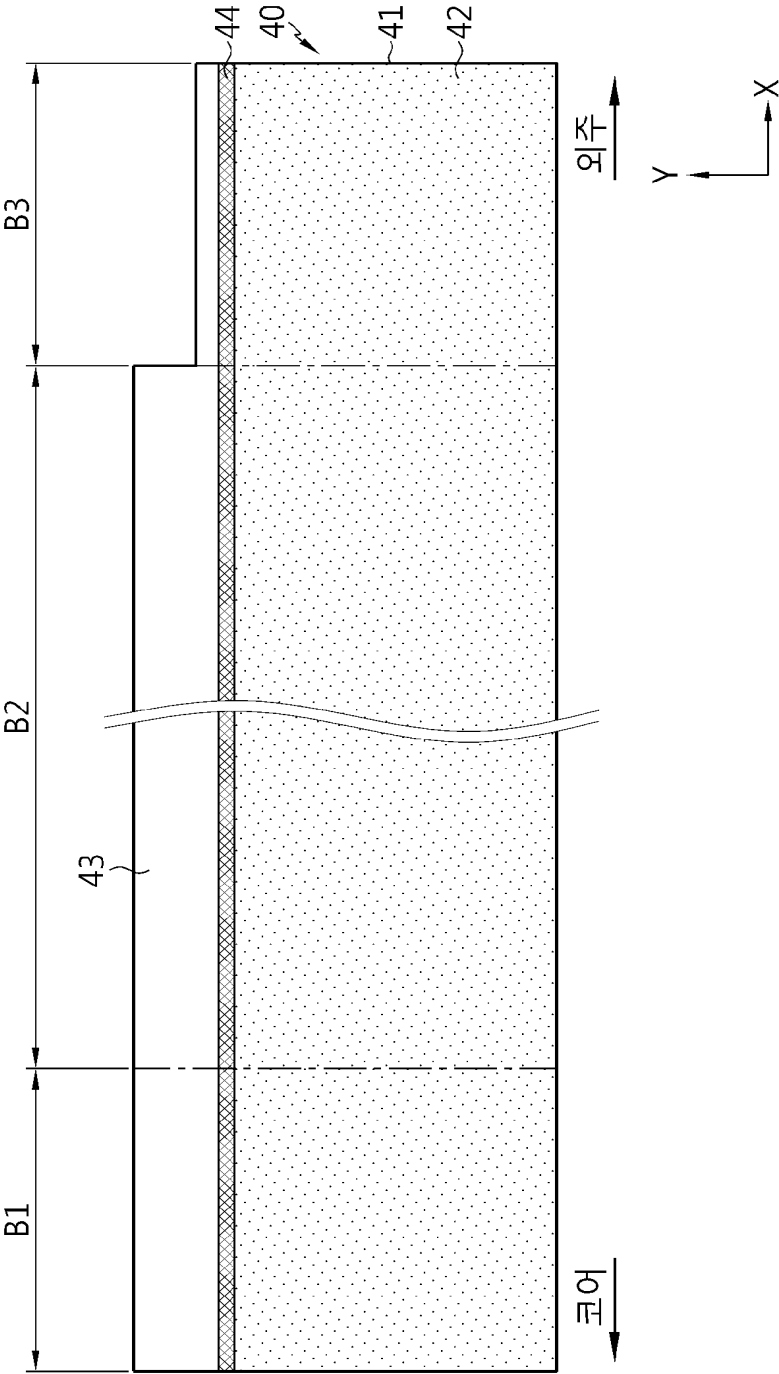


[도12b]

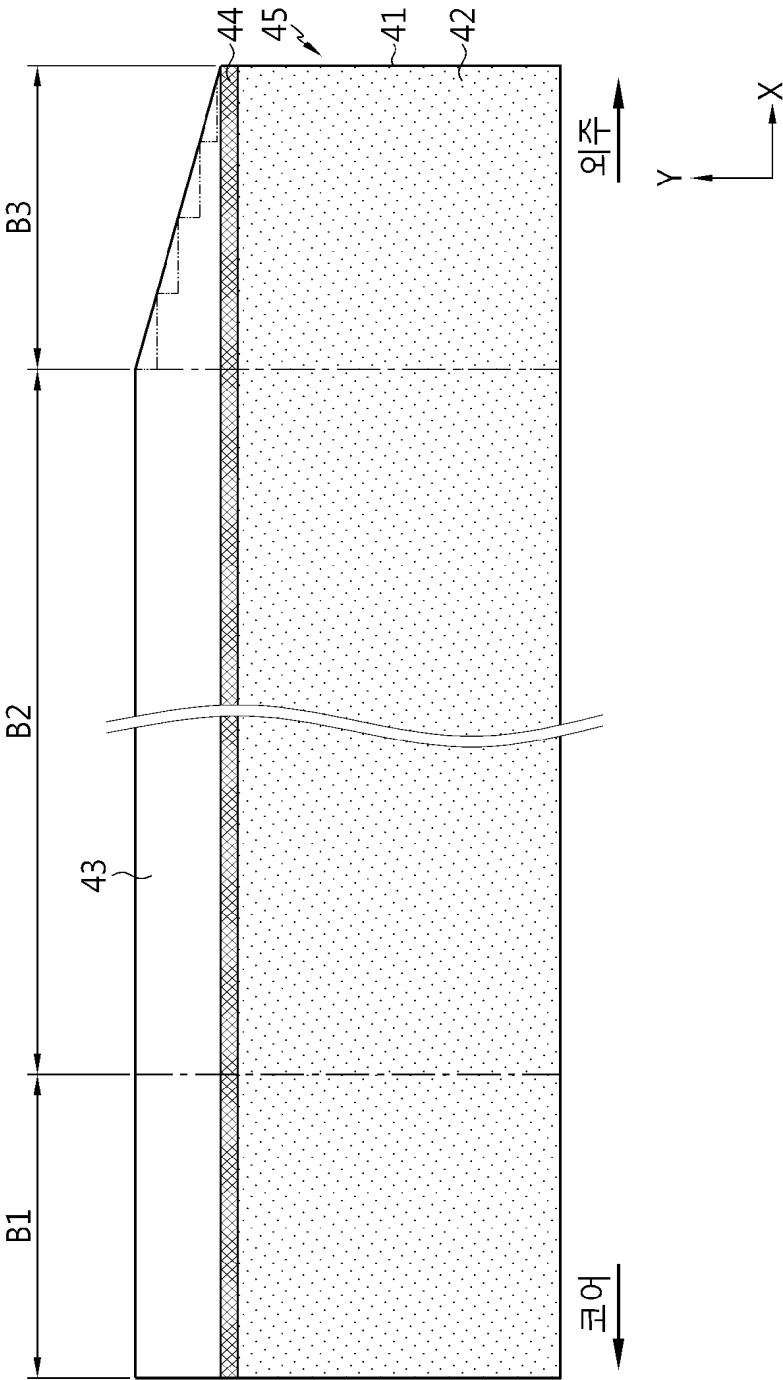
420 사이클 이후



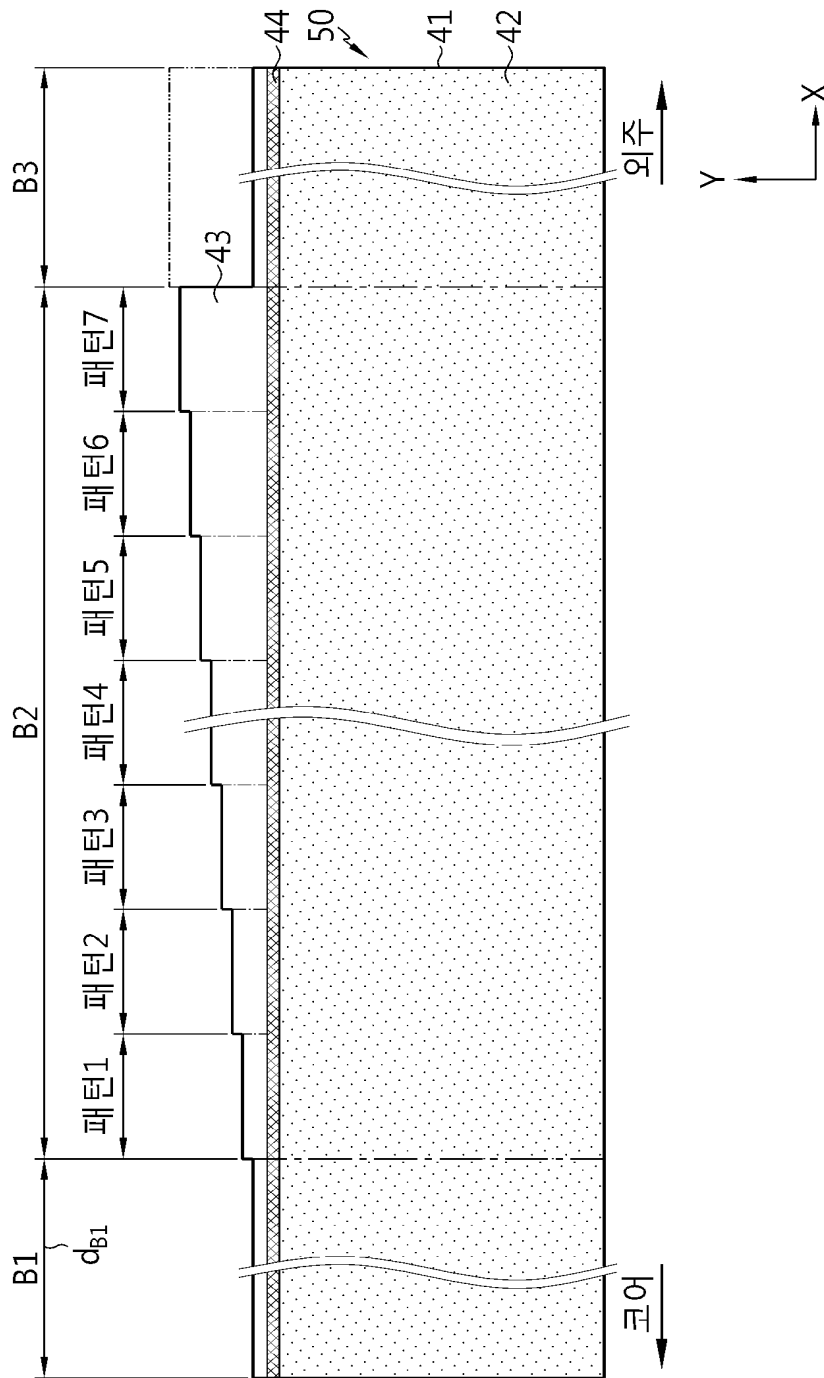
[도 14a]



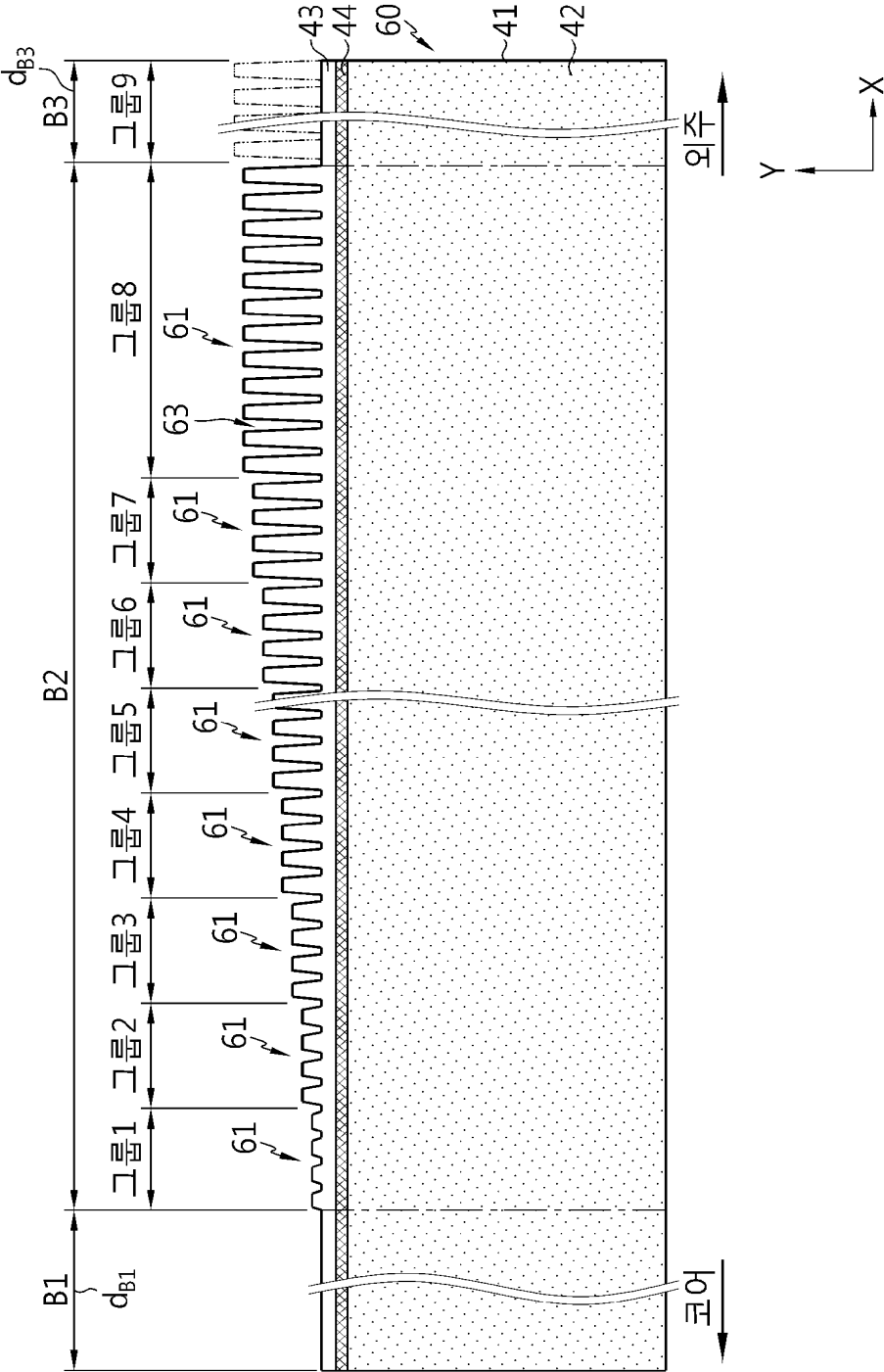
[도 14b]



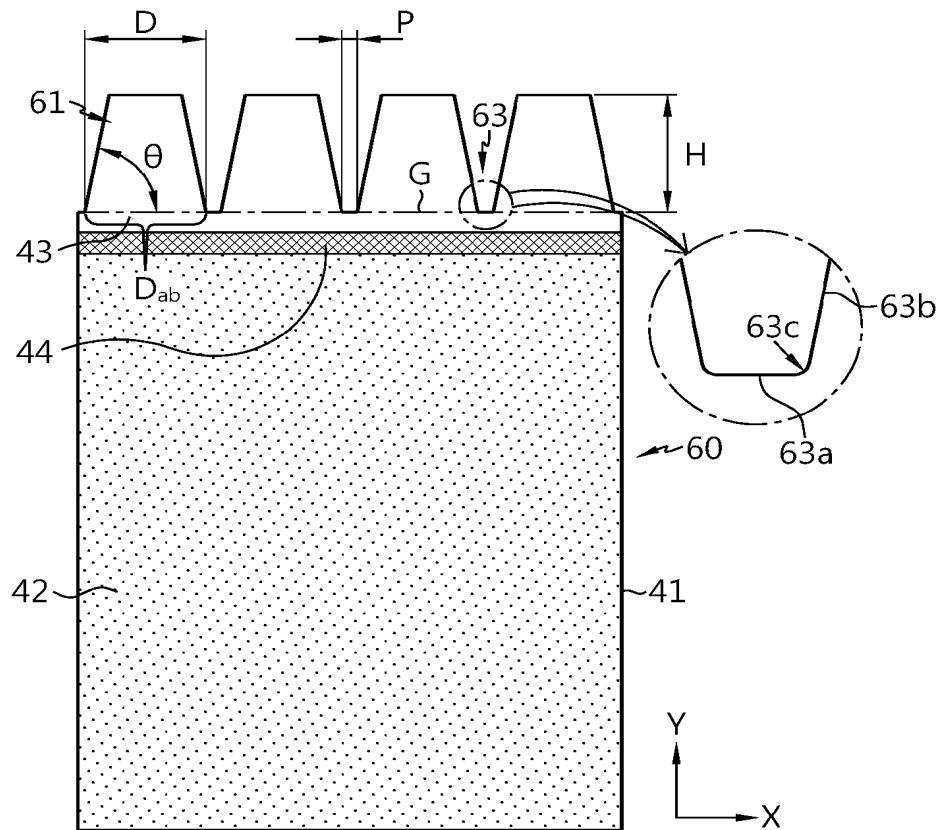
[도 14c]



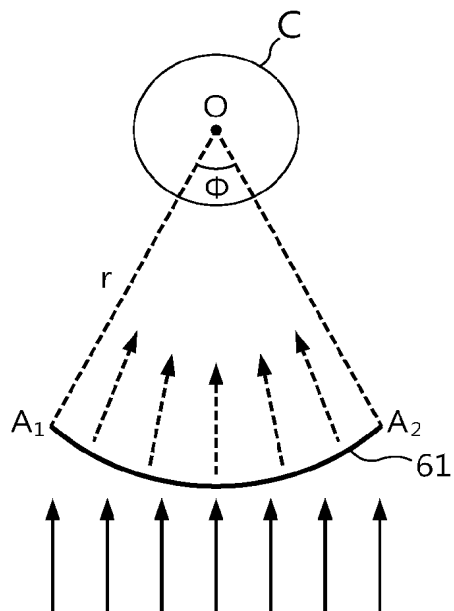
[도 14d]



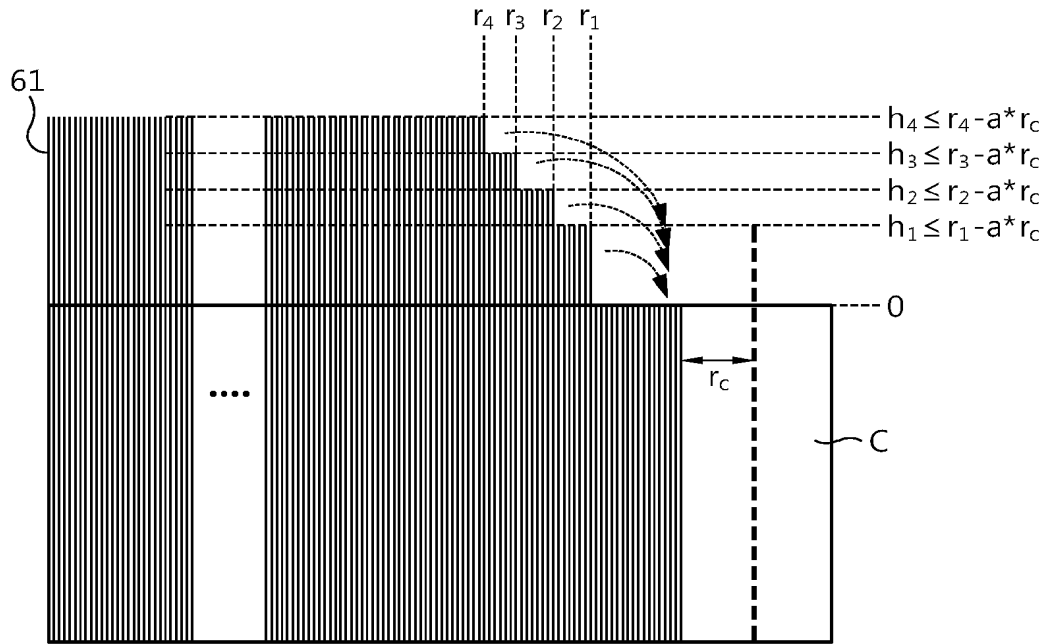
[도 14e]



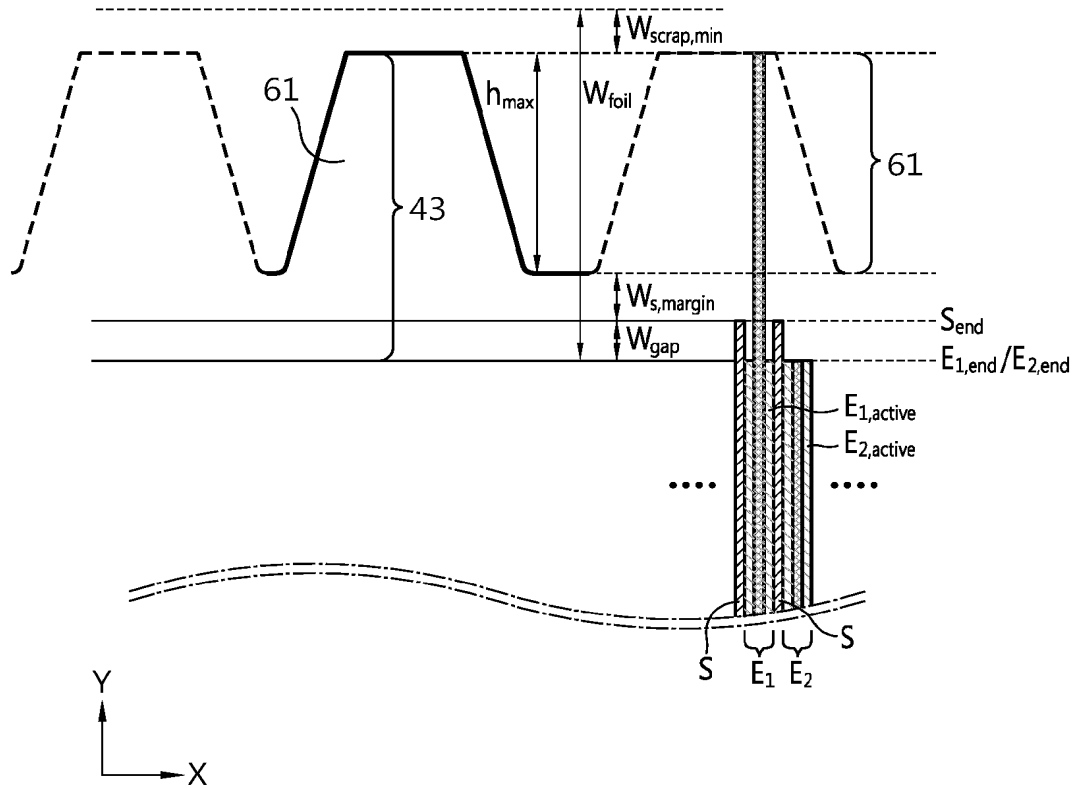
[도 14f]



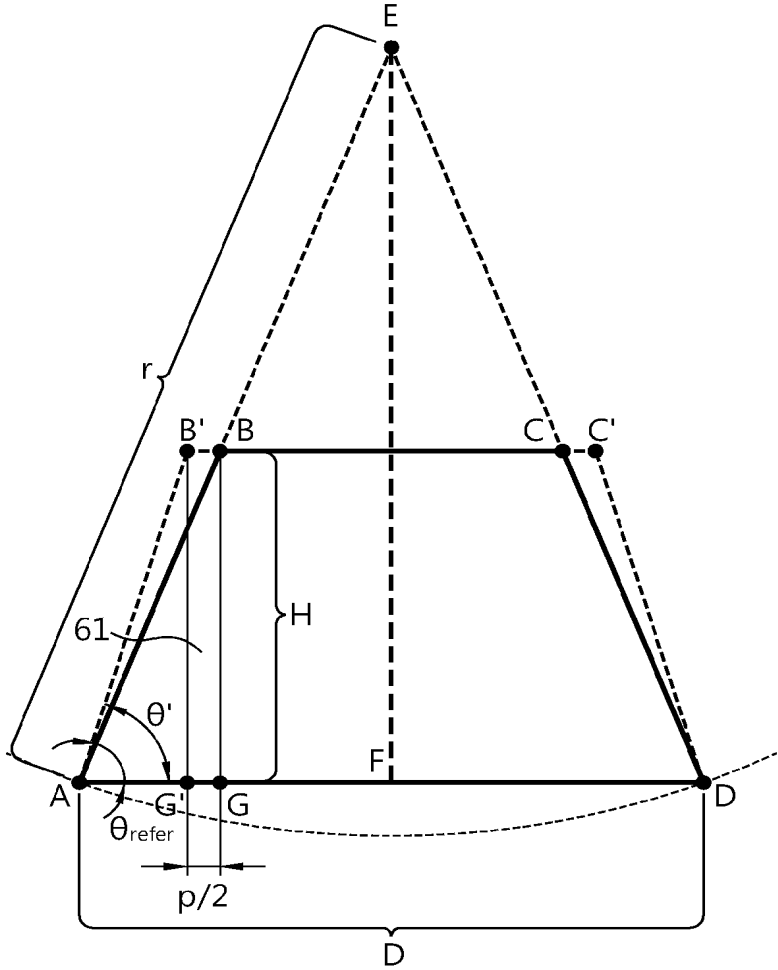
[도 14g]



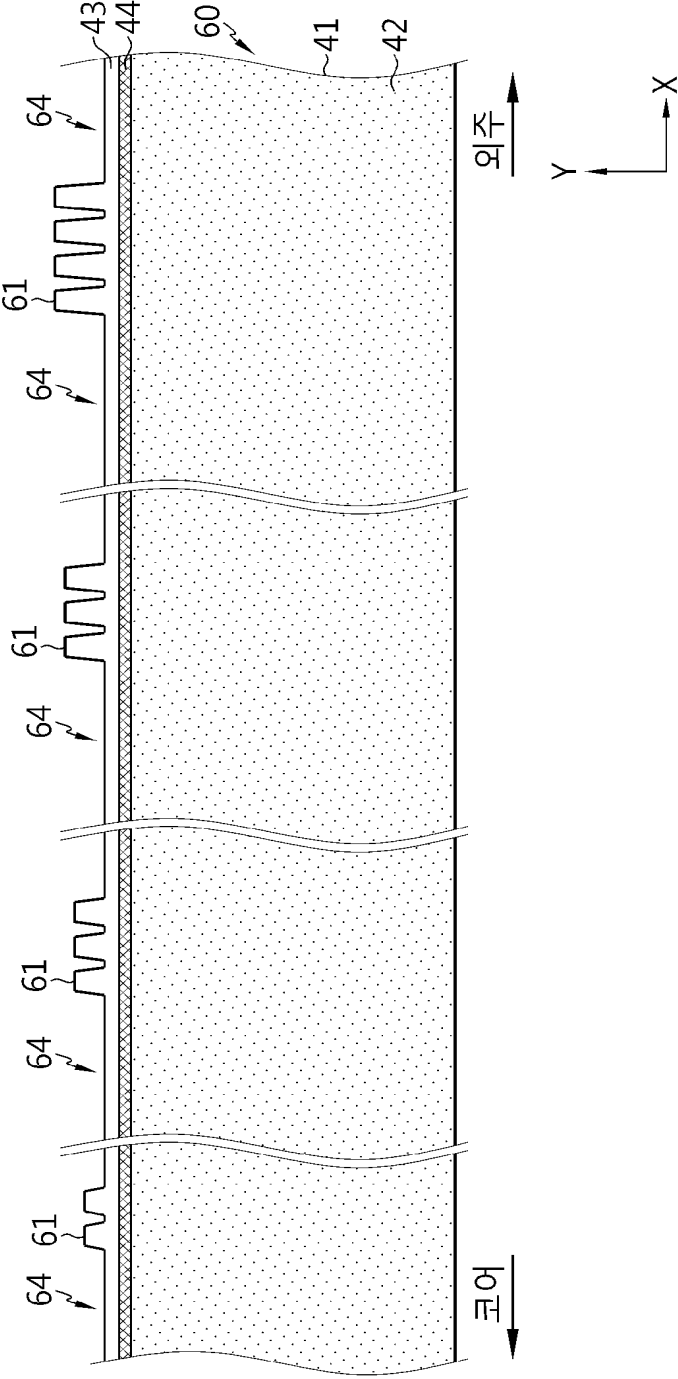
[도 14h]



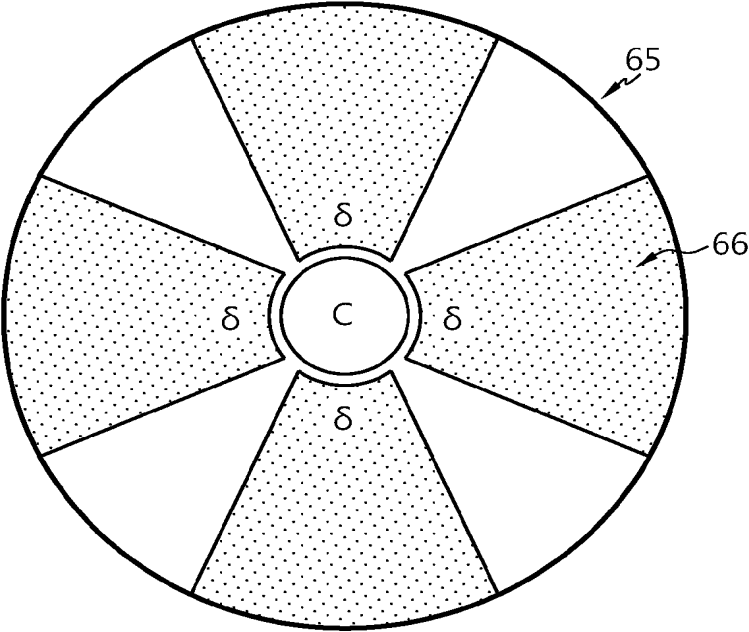
[도 14i]



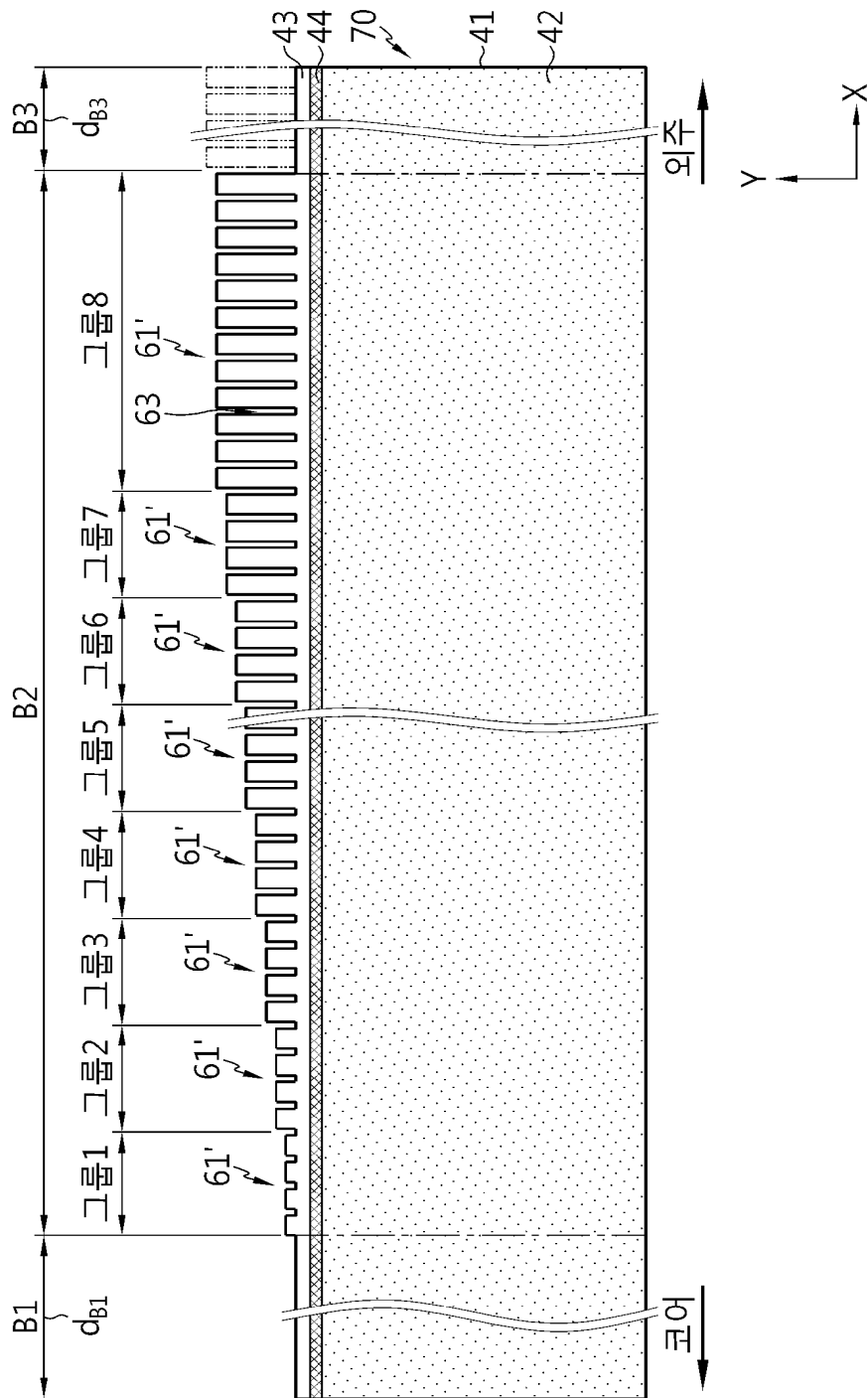
[도 14j]



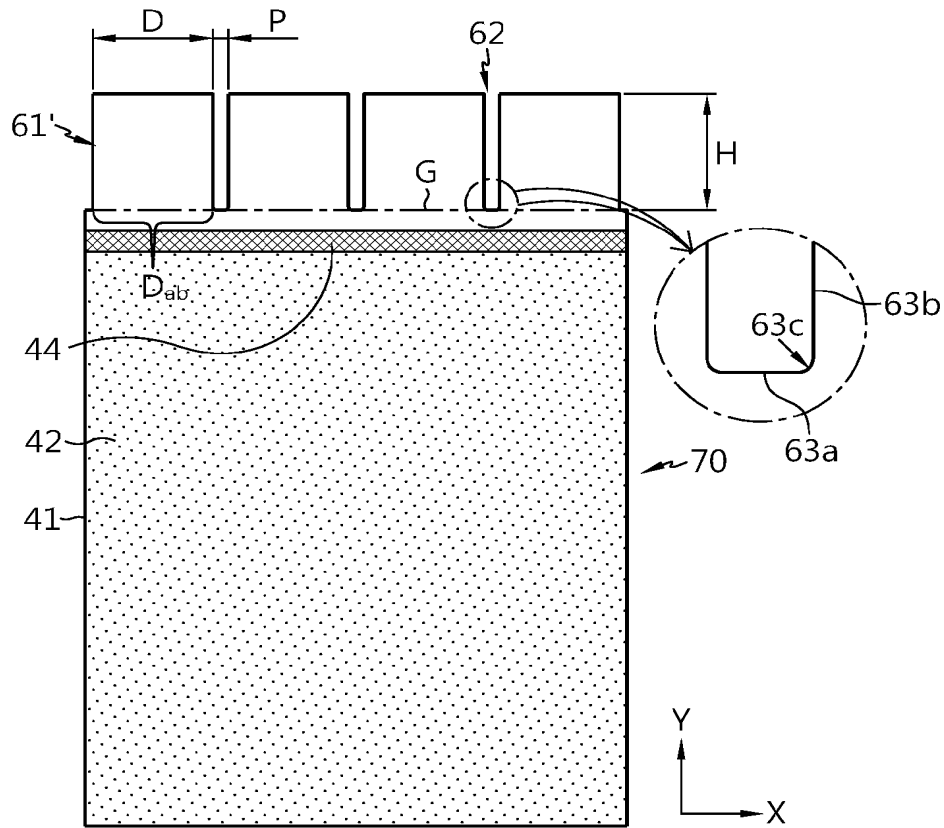
[도 14k]



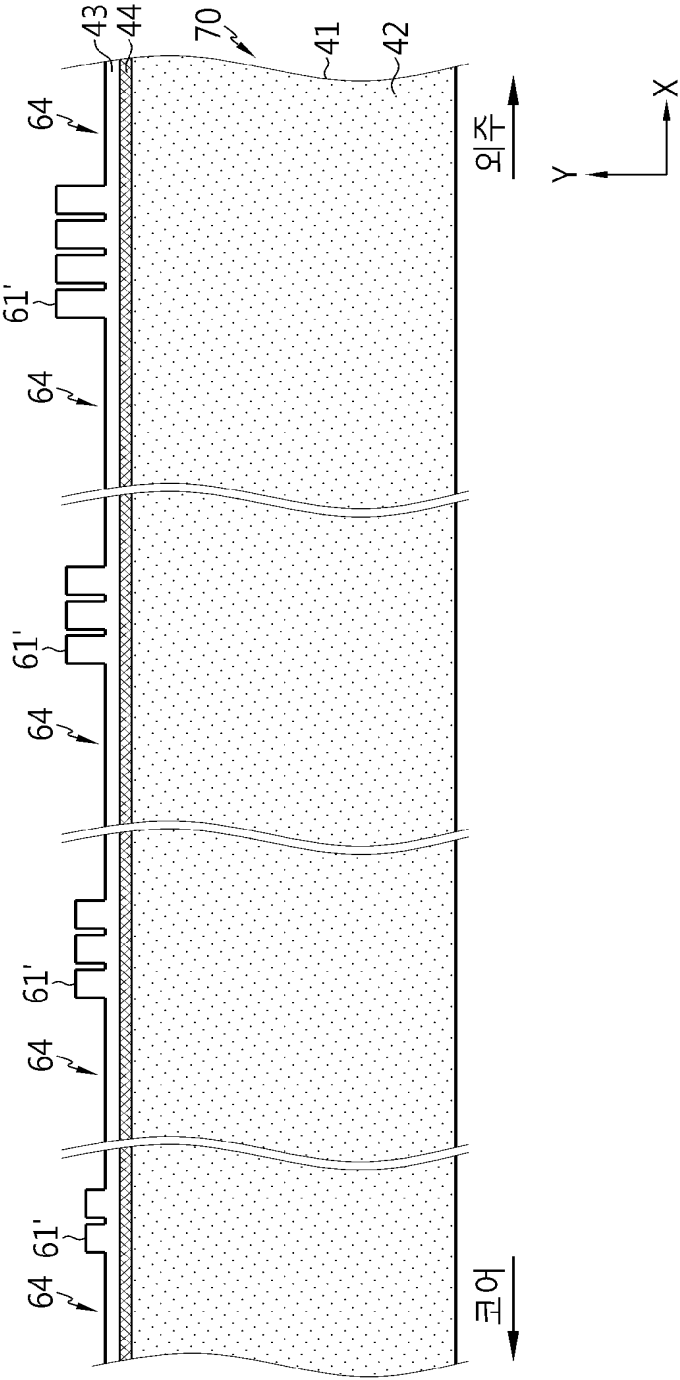
[도 15a]



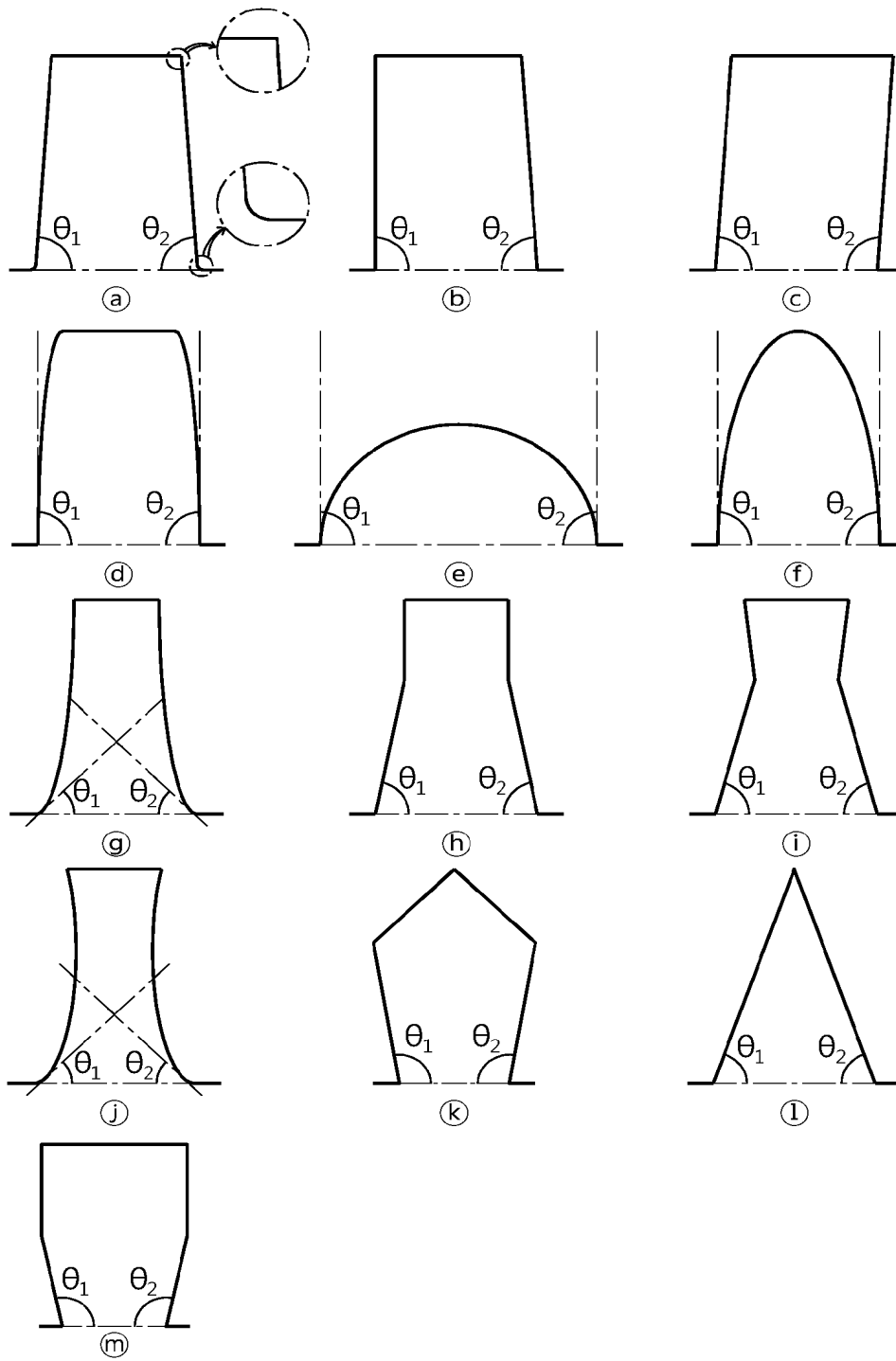
[도 15b]



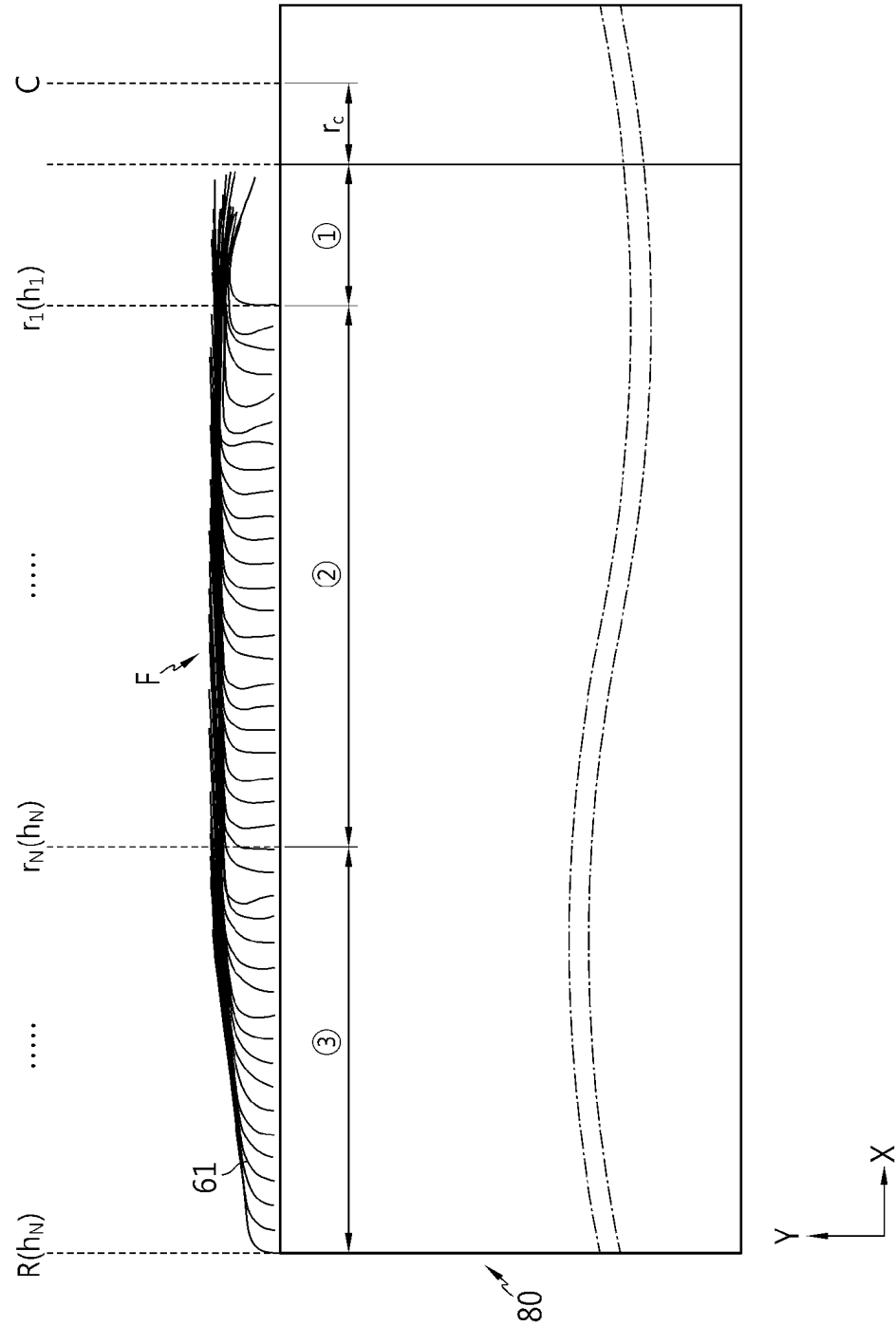
[도15c]



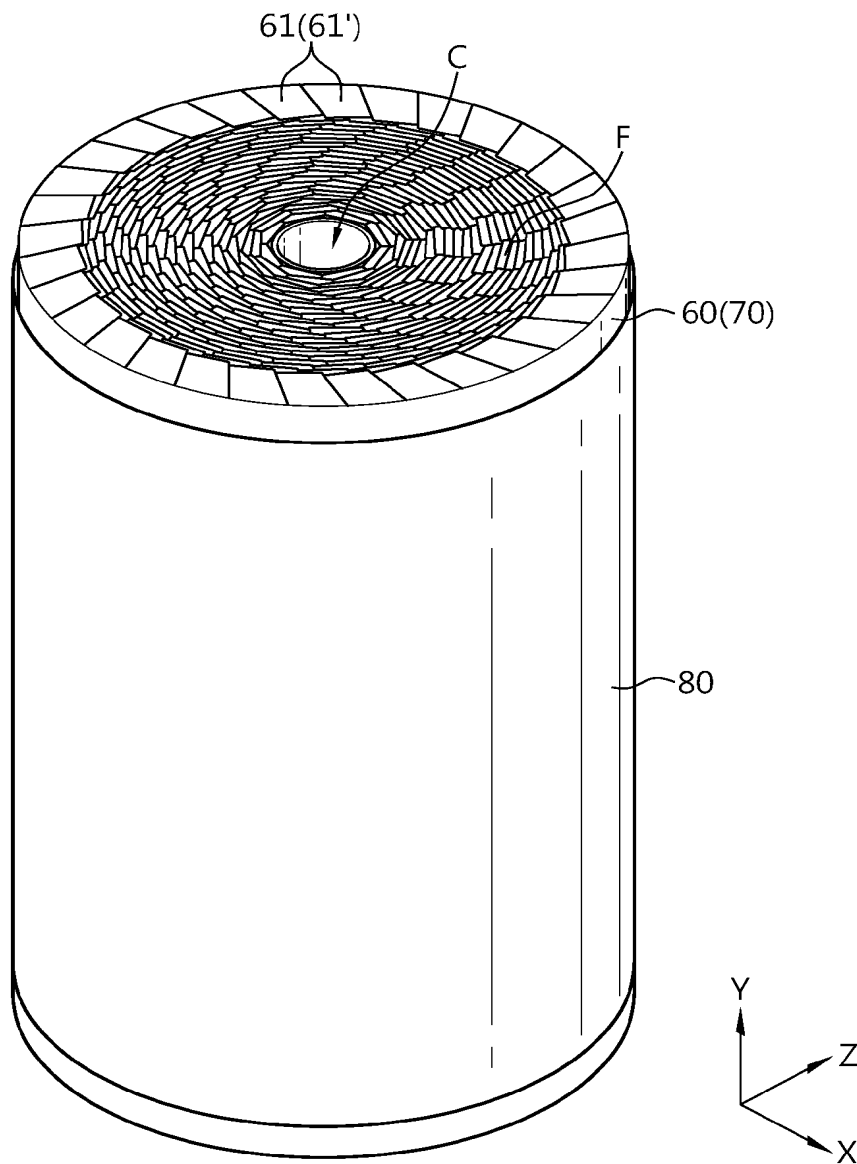
[도16]



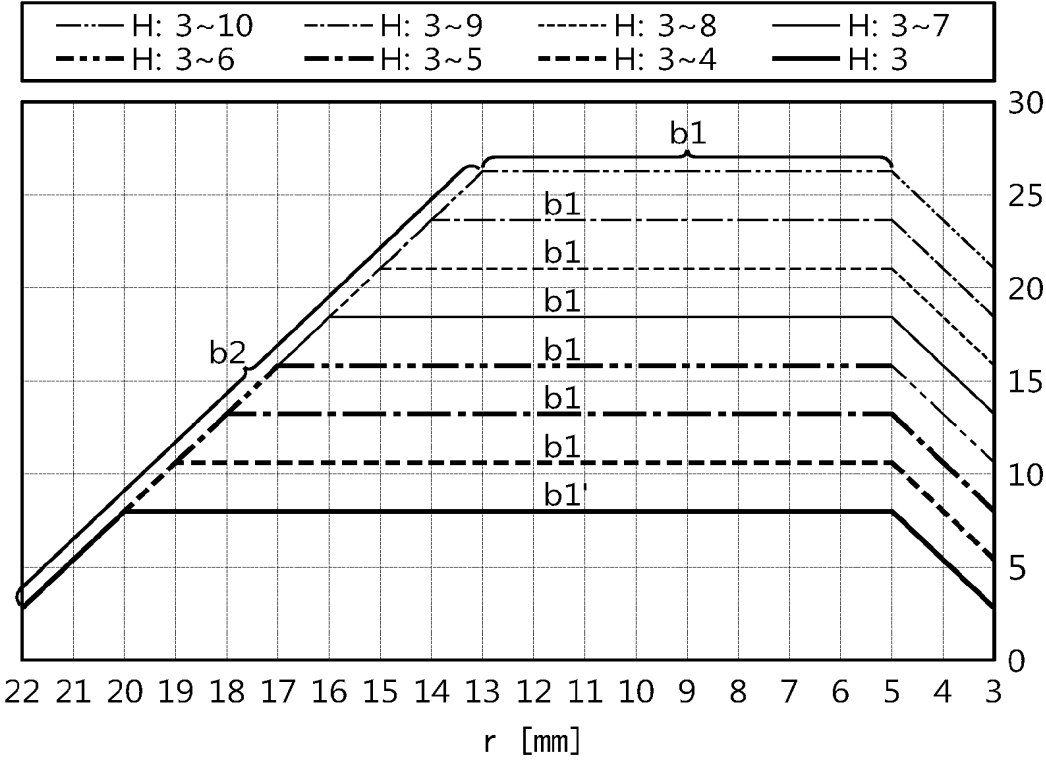
[도 17a]



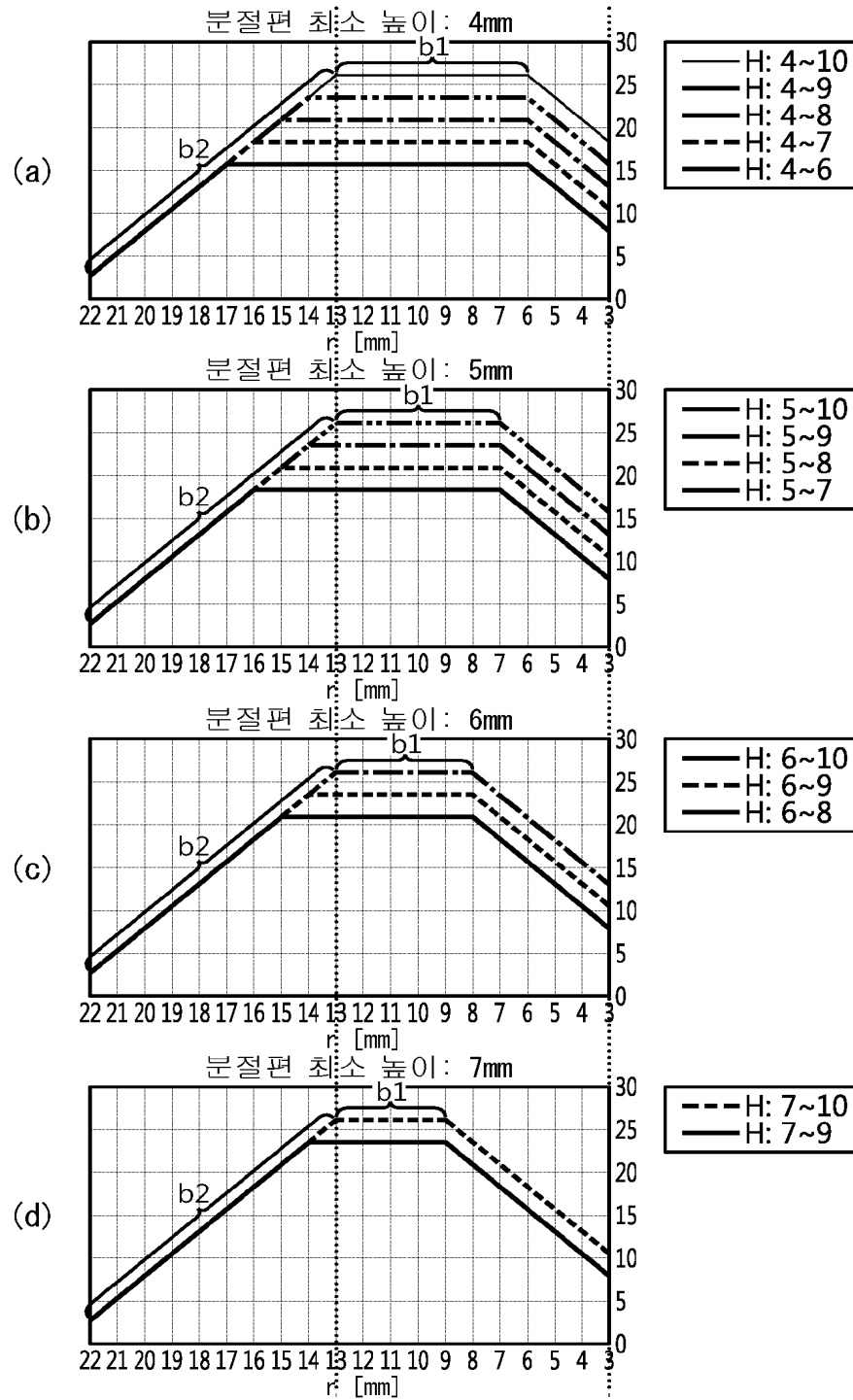
[도17b]



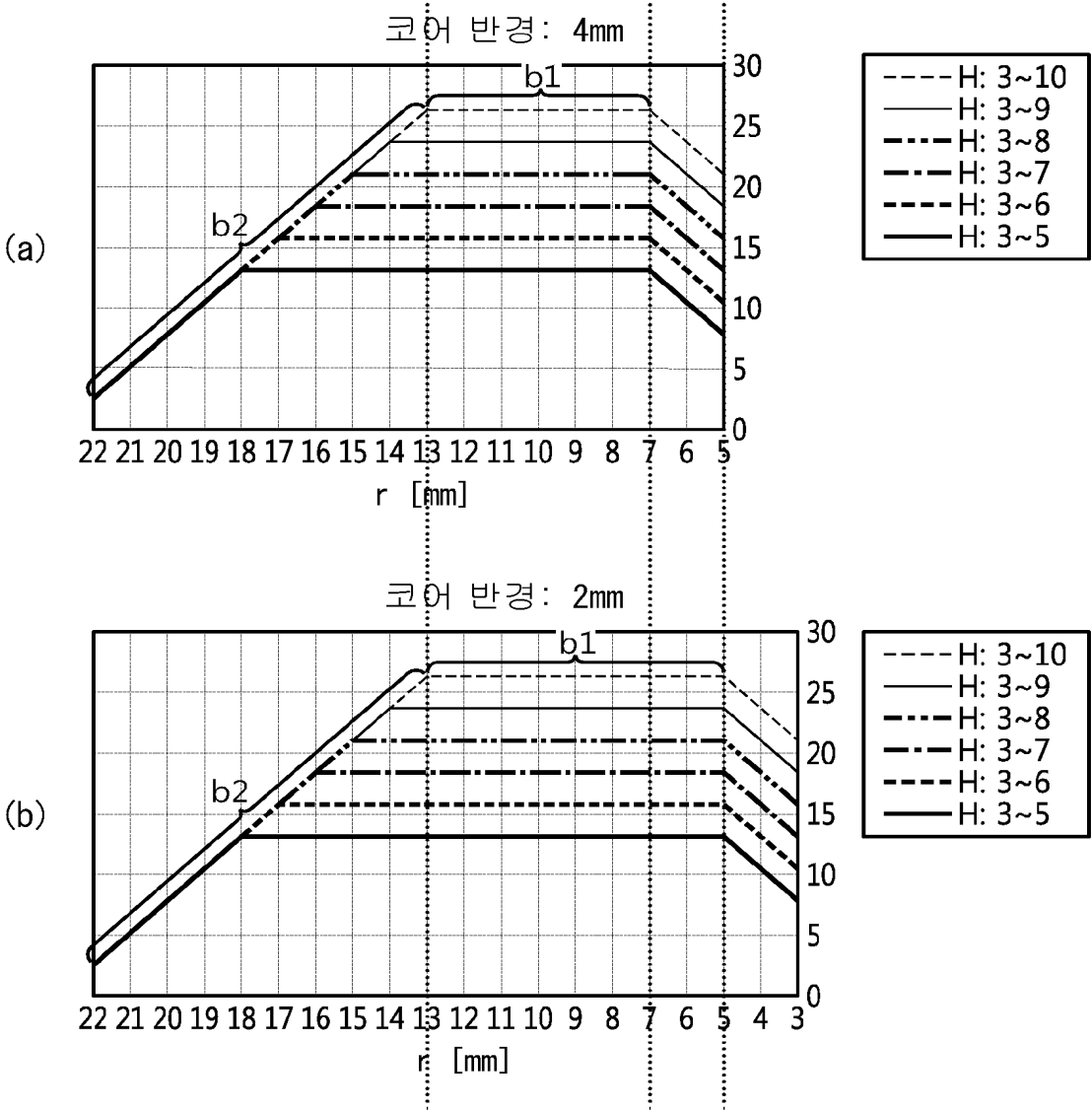
[도17c]



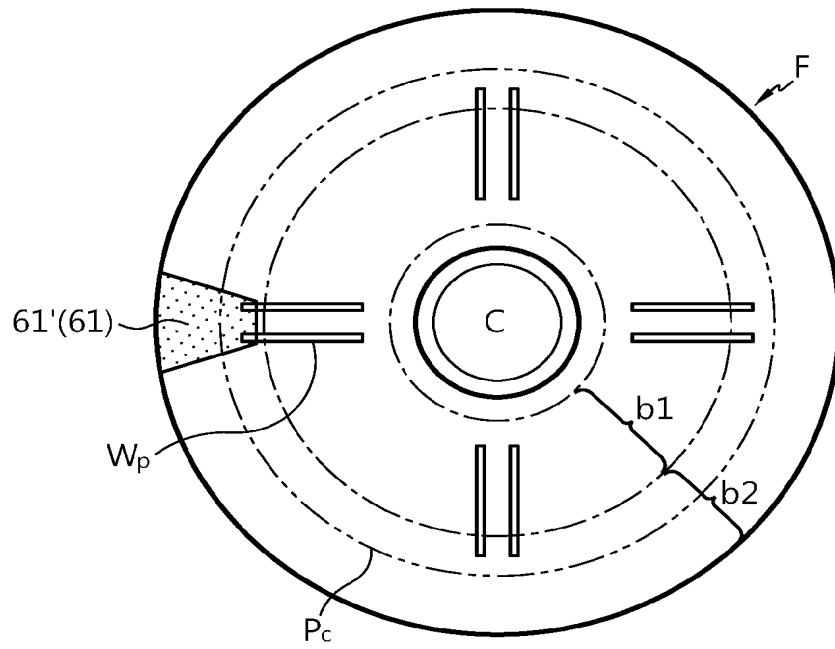
[도17d]



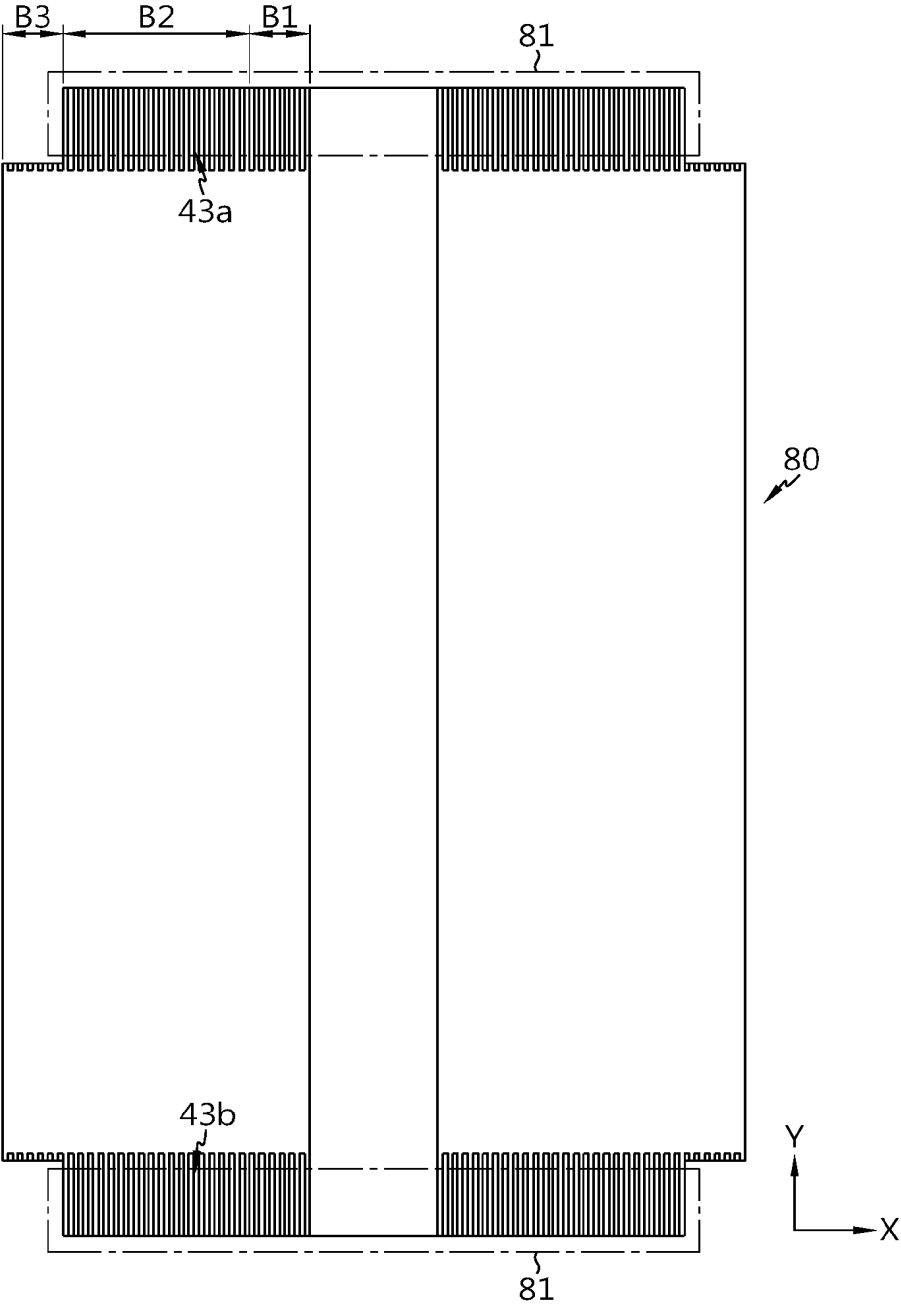
[도17e]



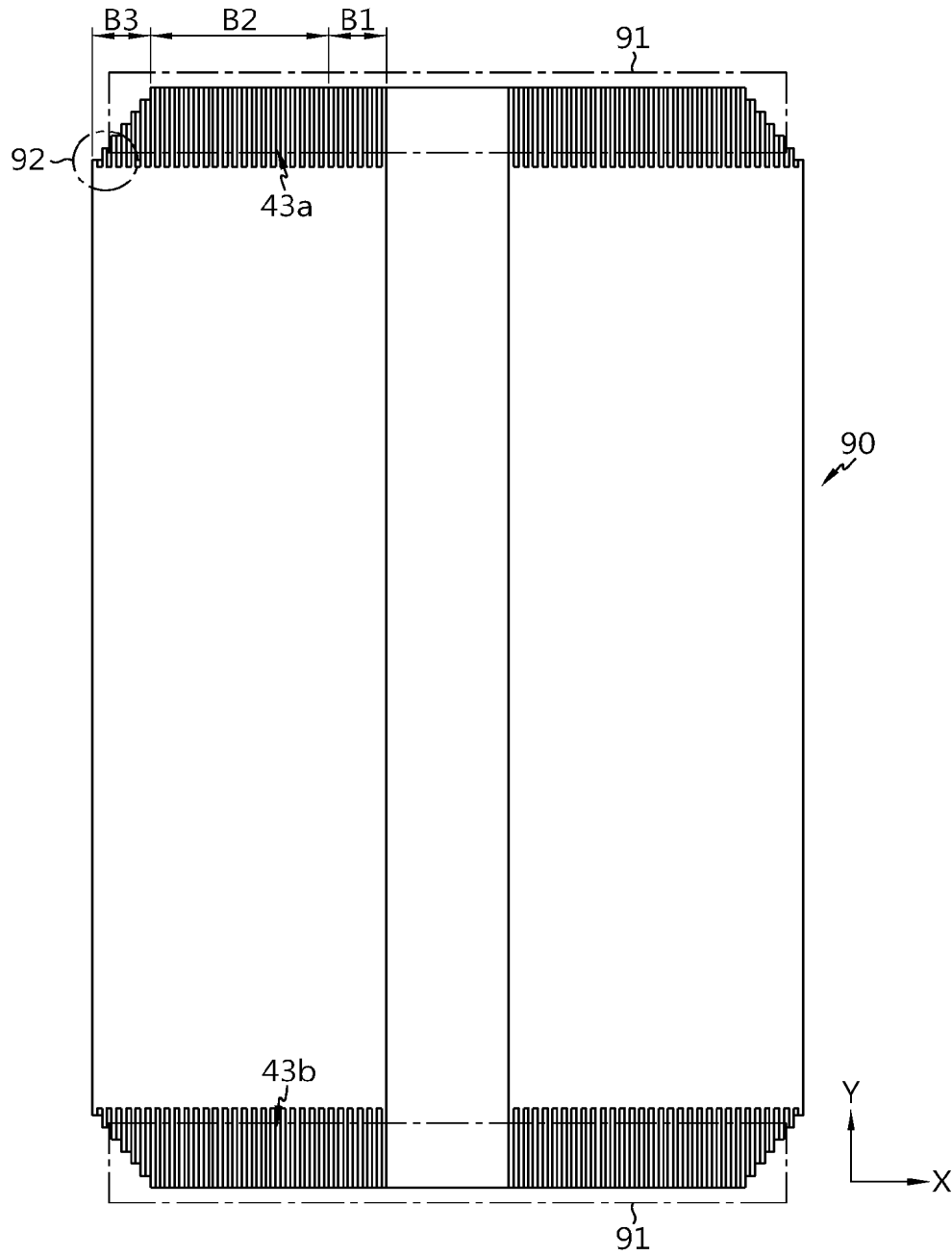
[도 17f]



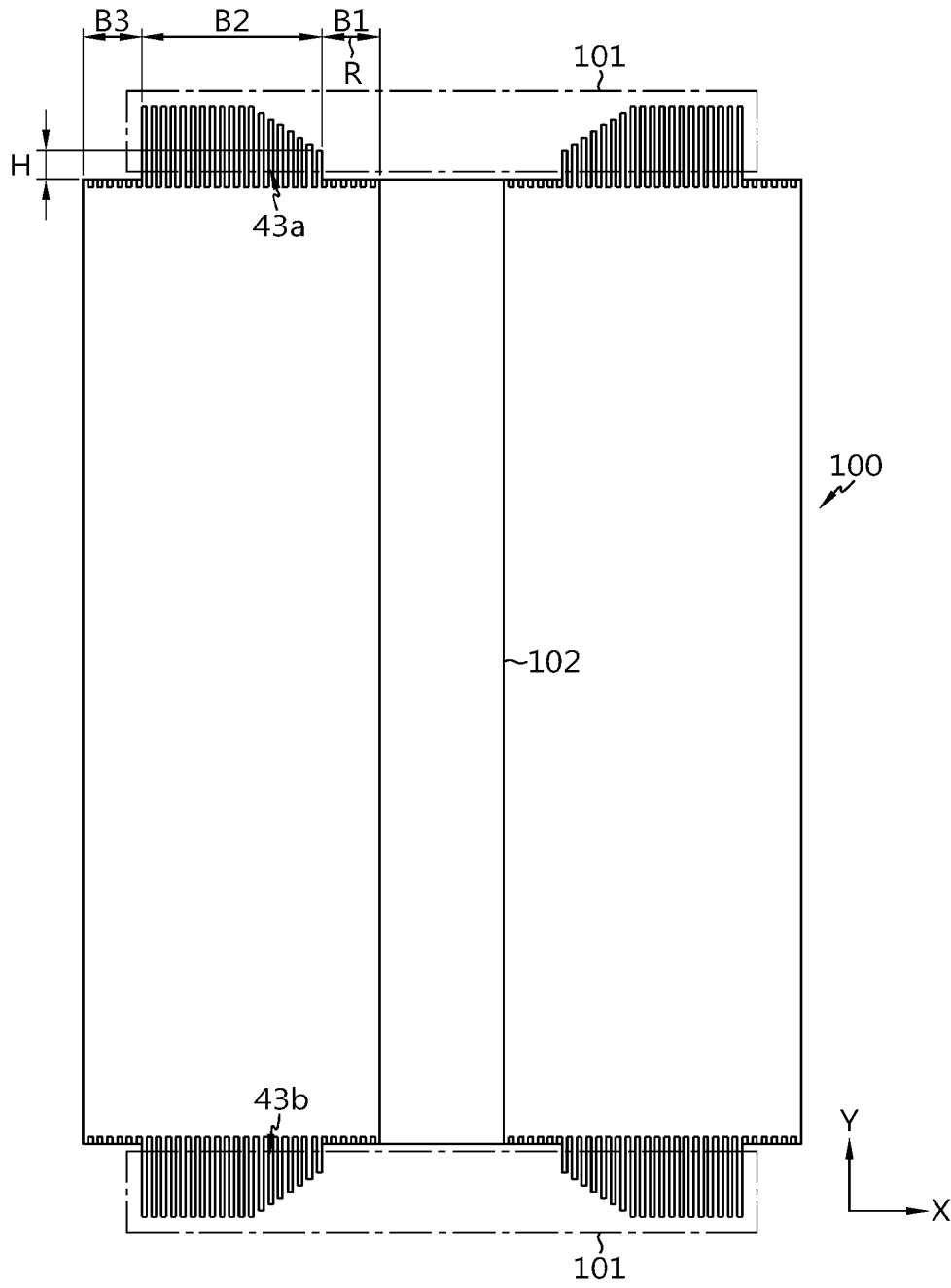
[도18]



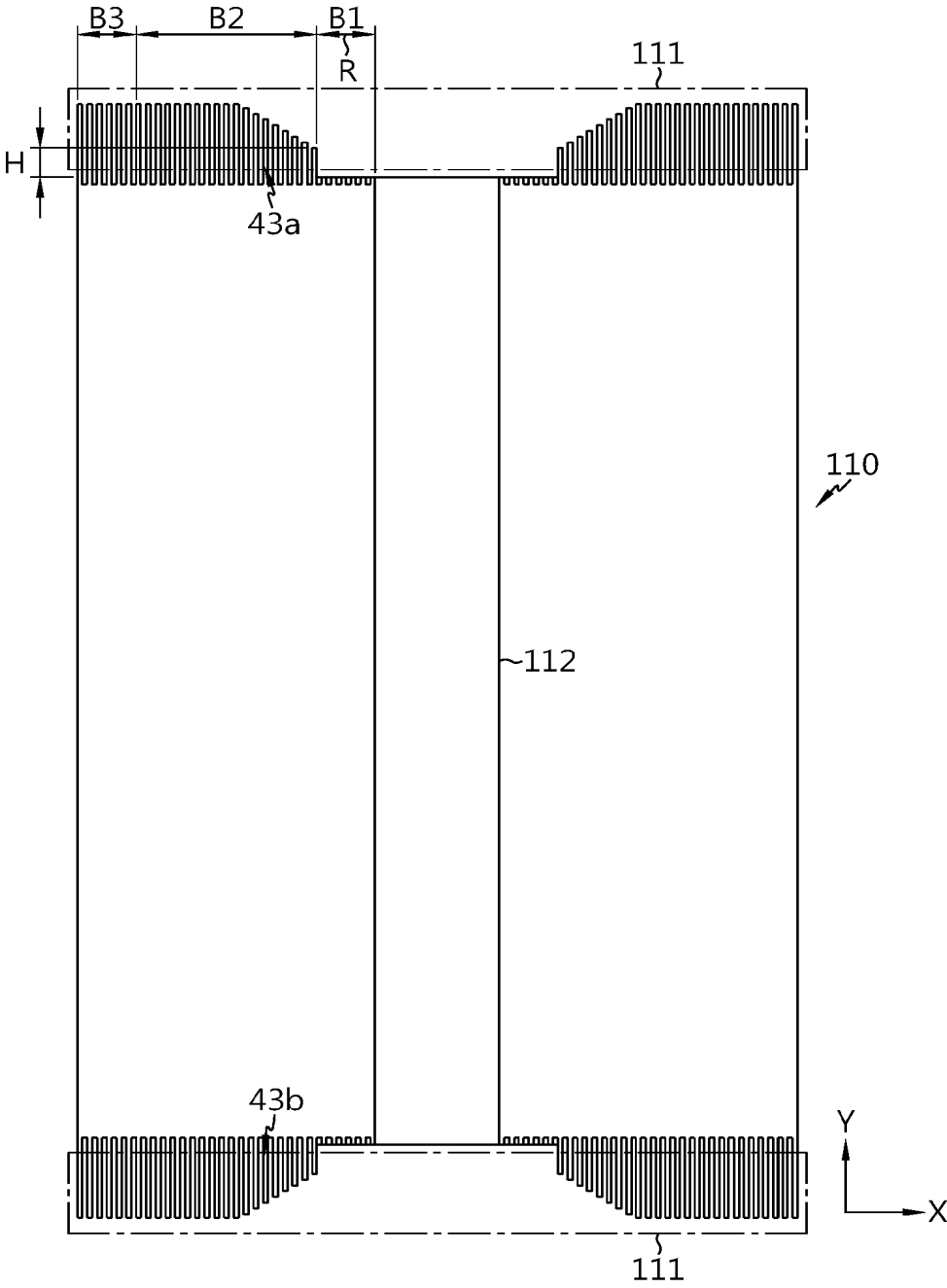
[도 19]



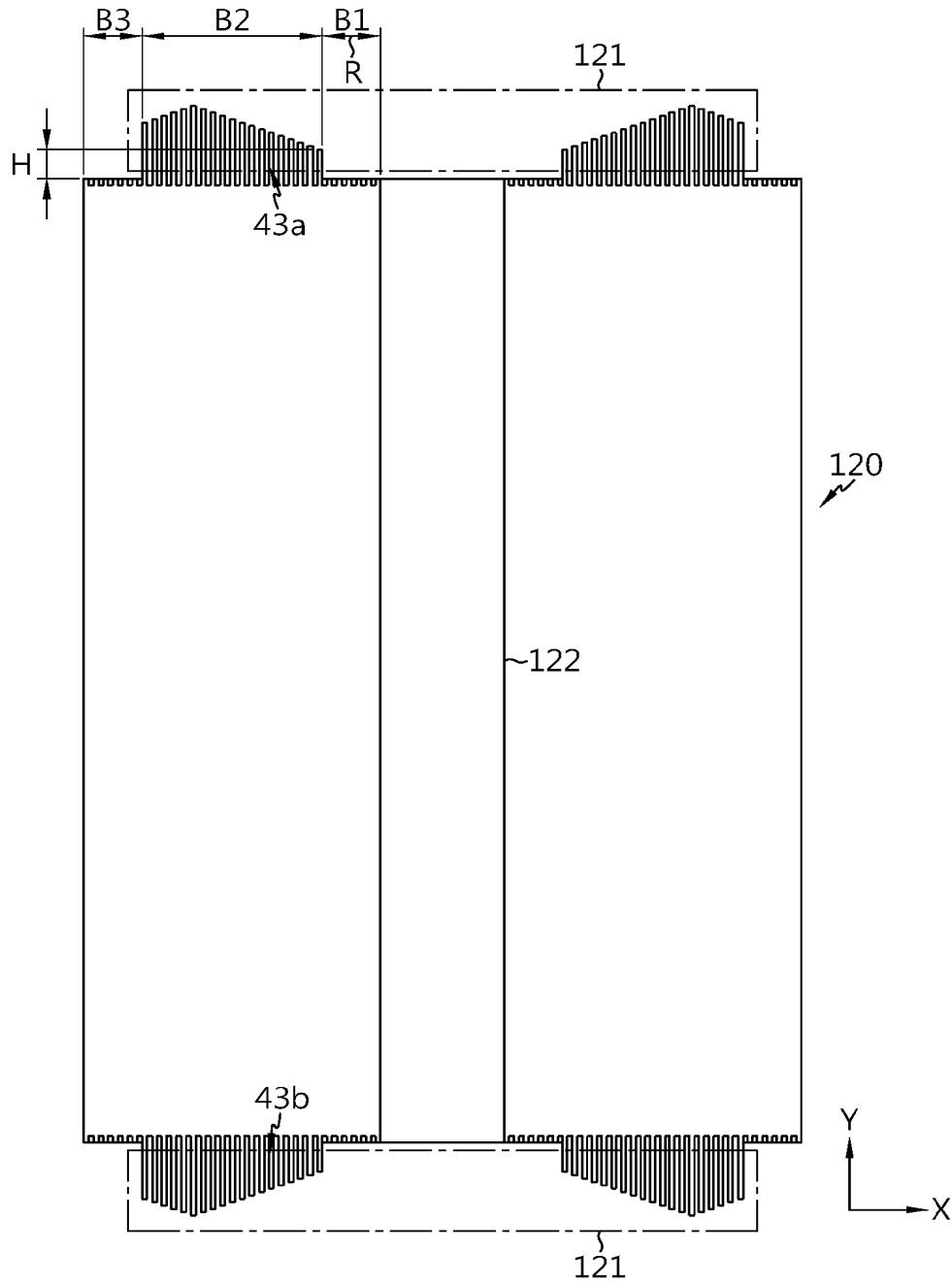
[도20]



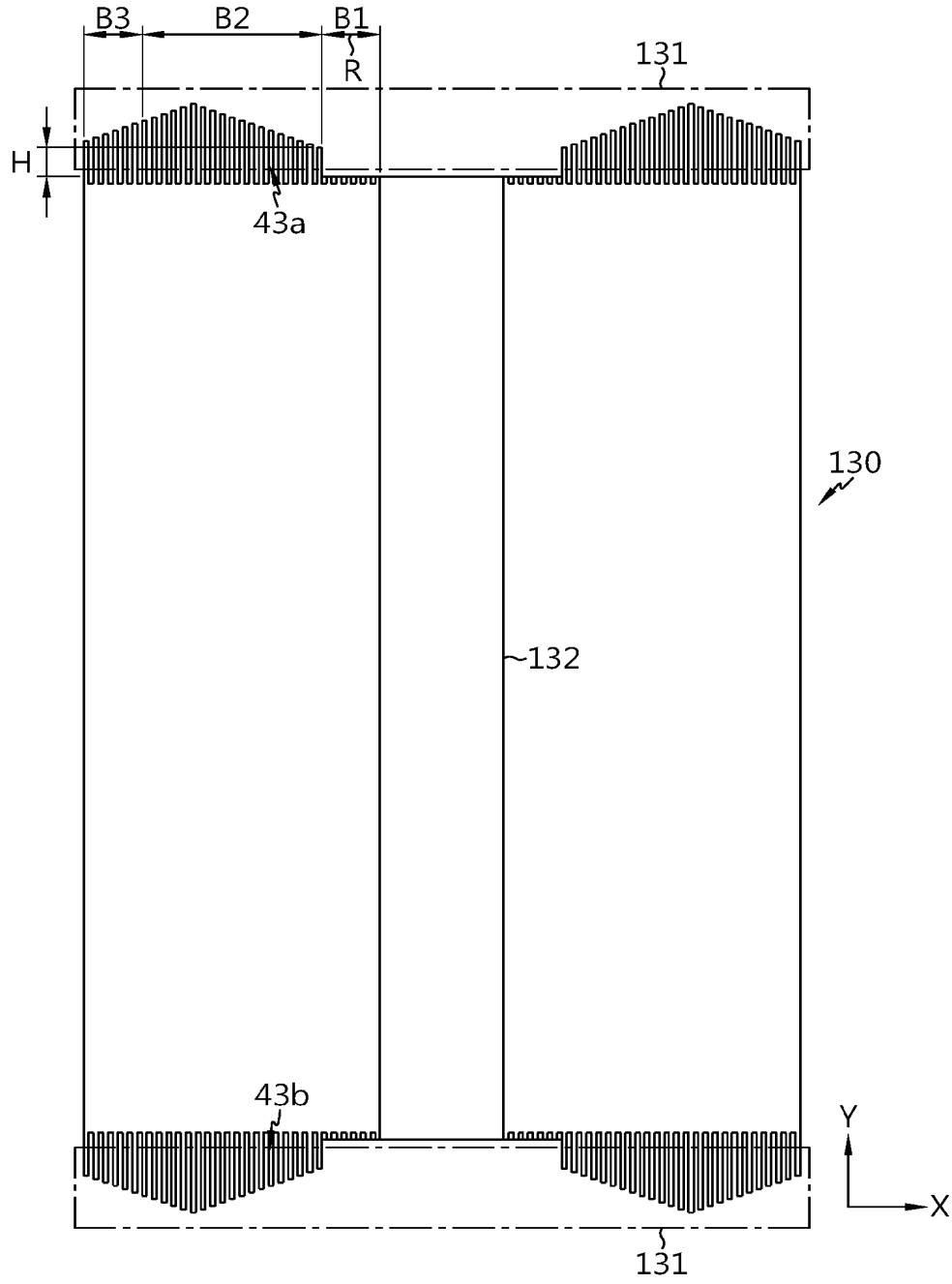
[도21]



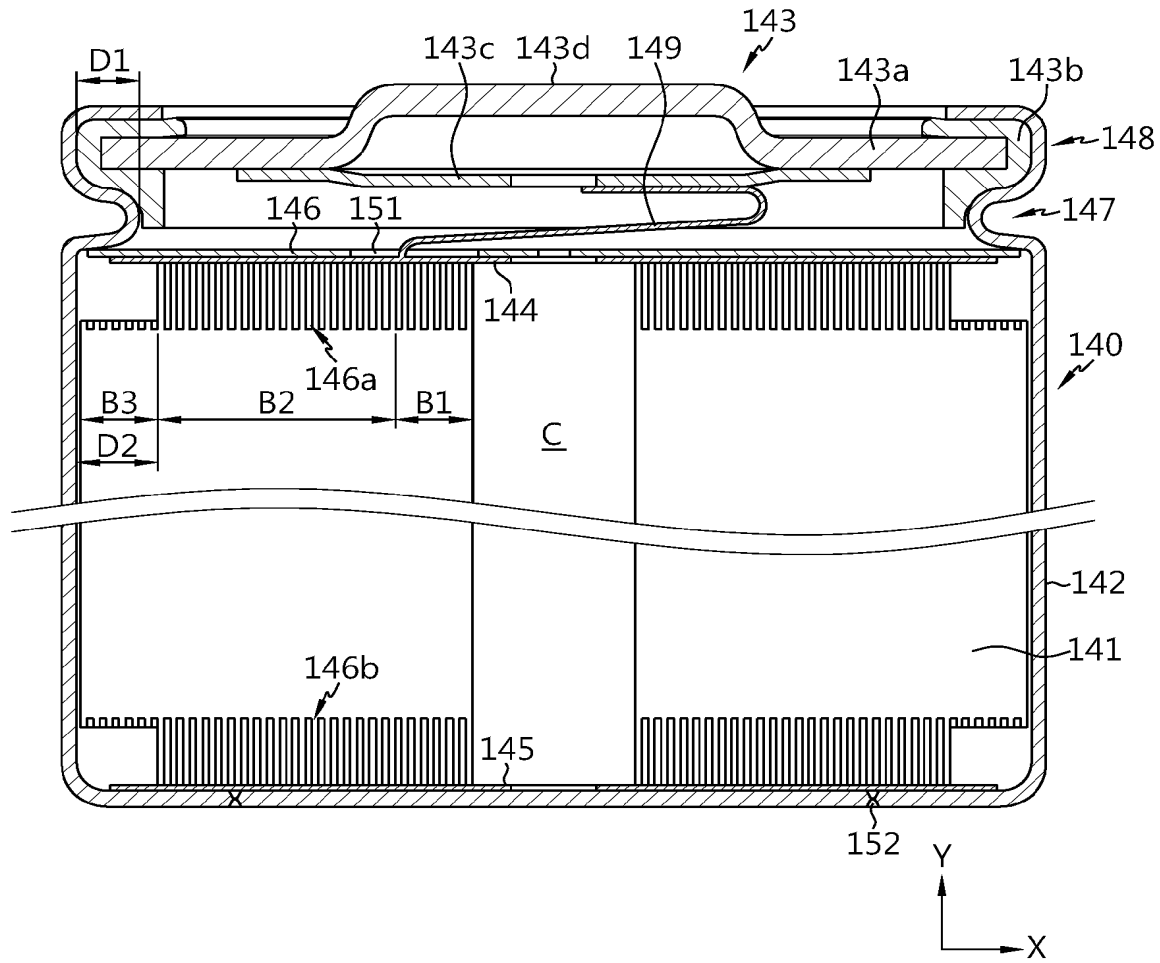
[도22]



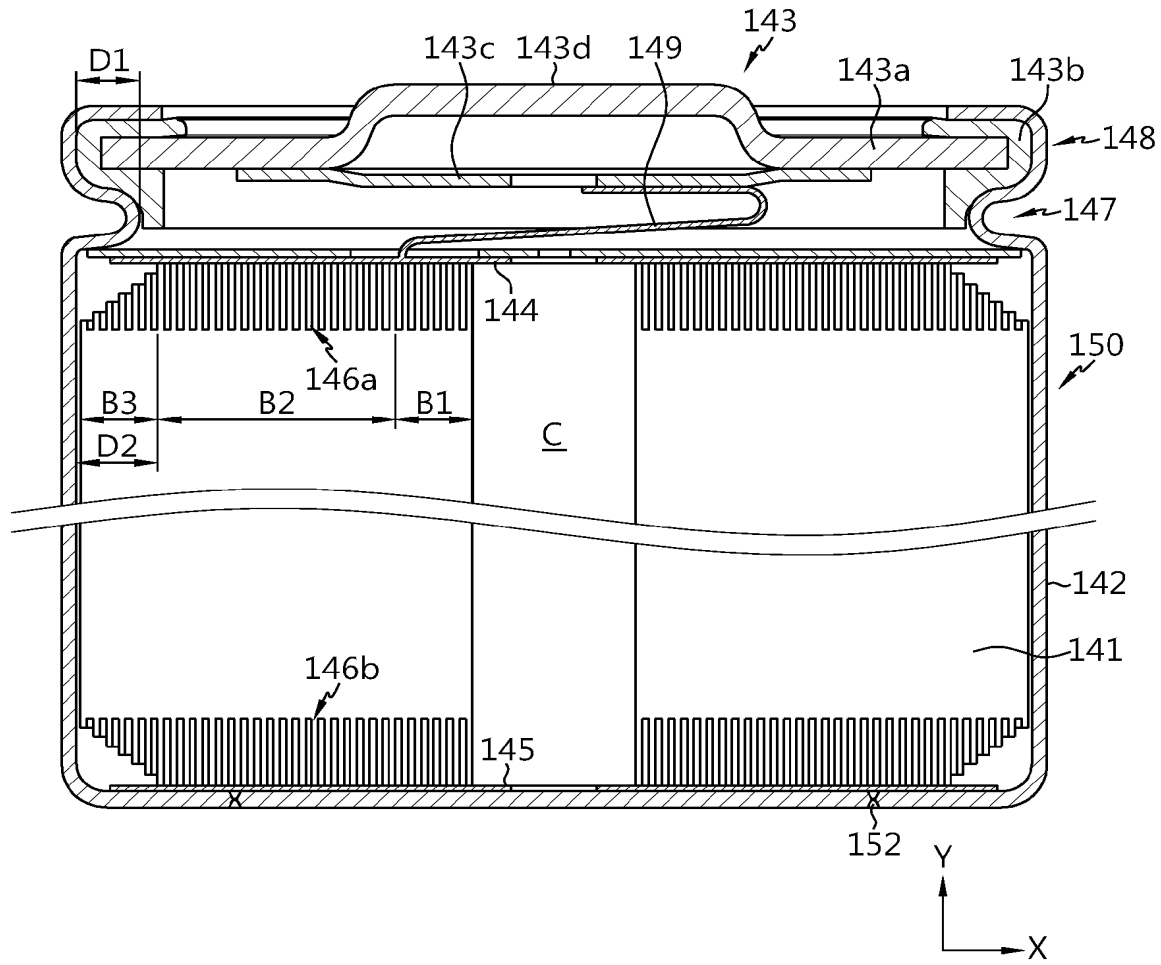
[도23]



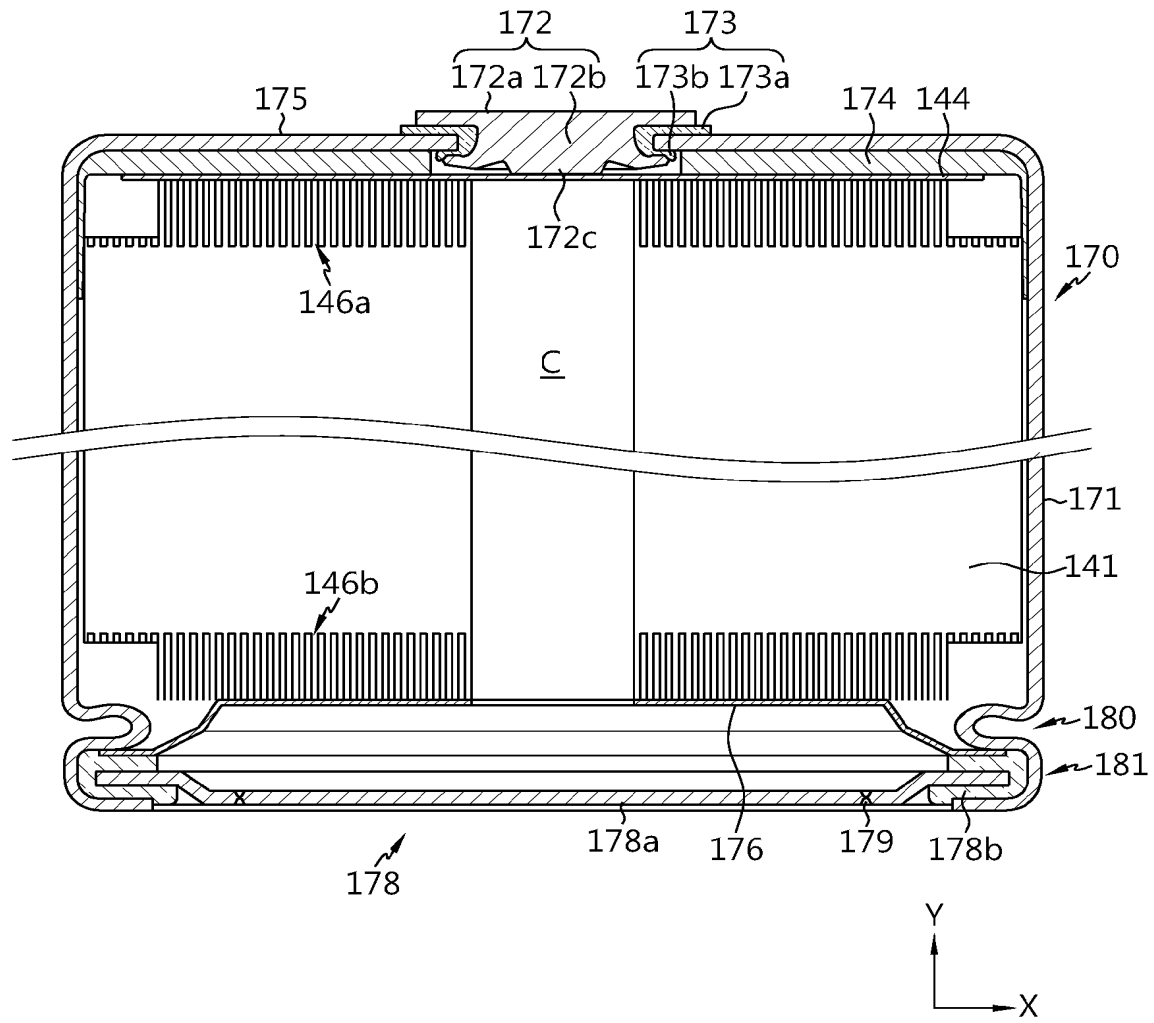
[도 24]



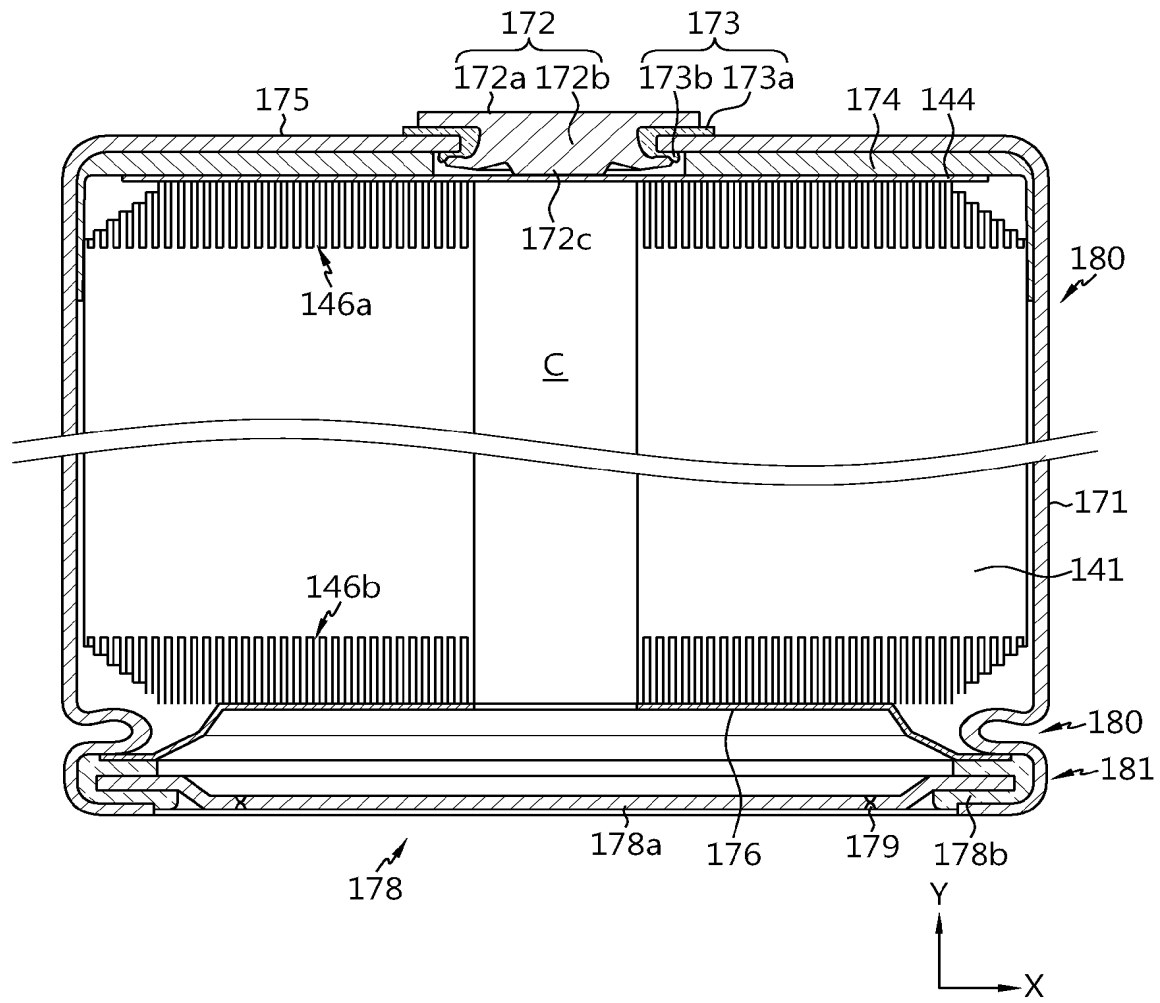
[도25]



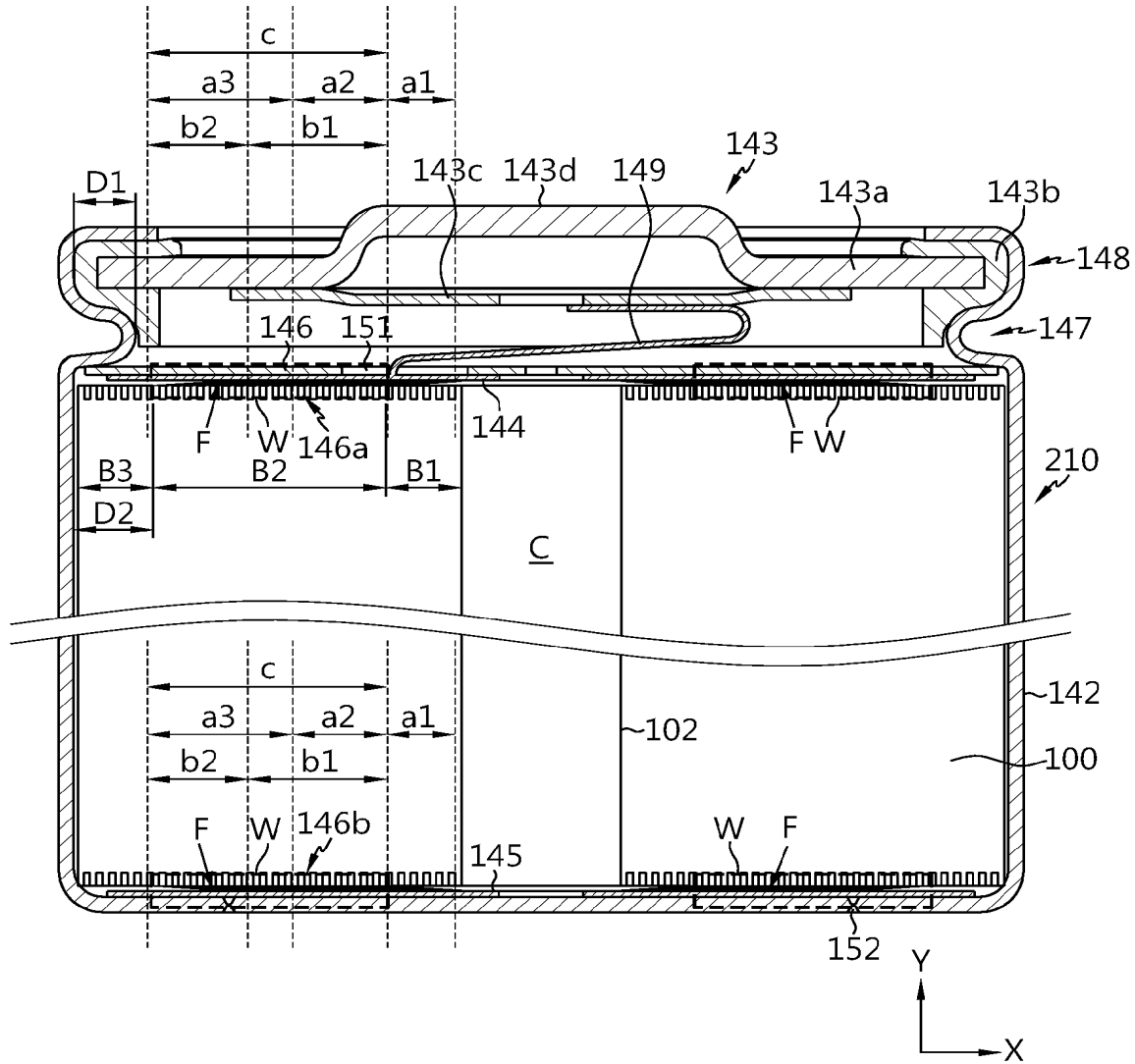
[도27]



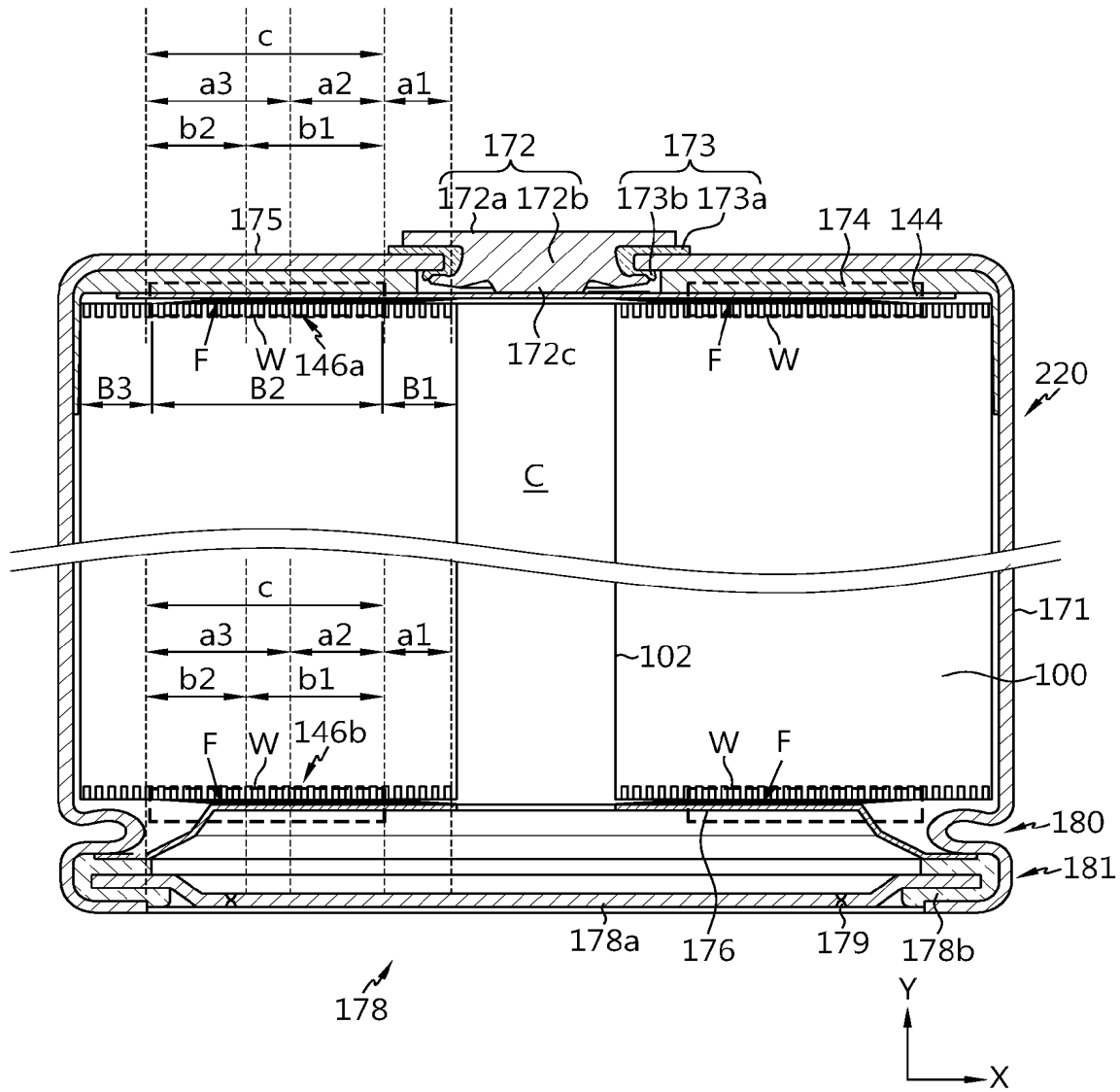
[도28]



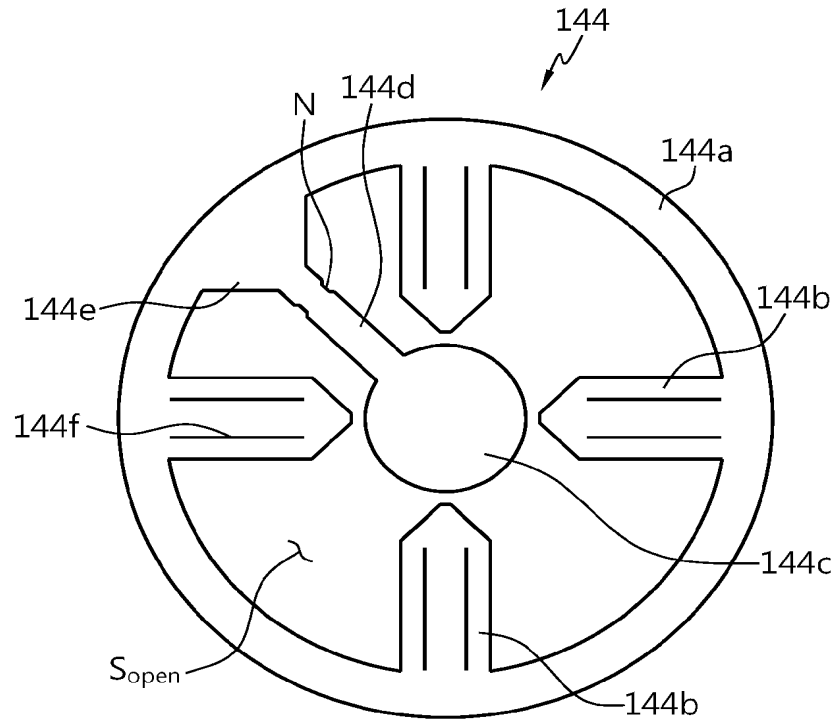
[도31]



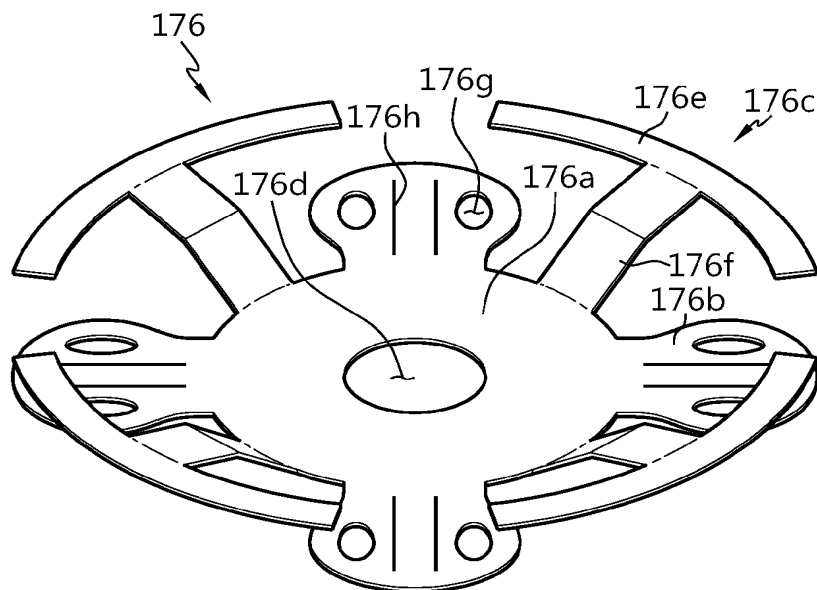
[도32]



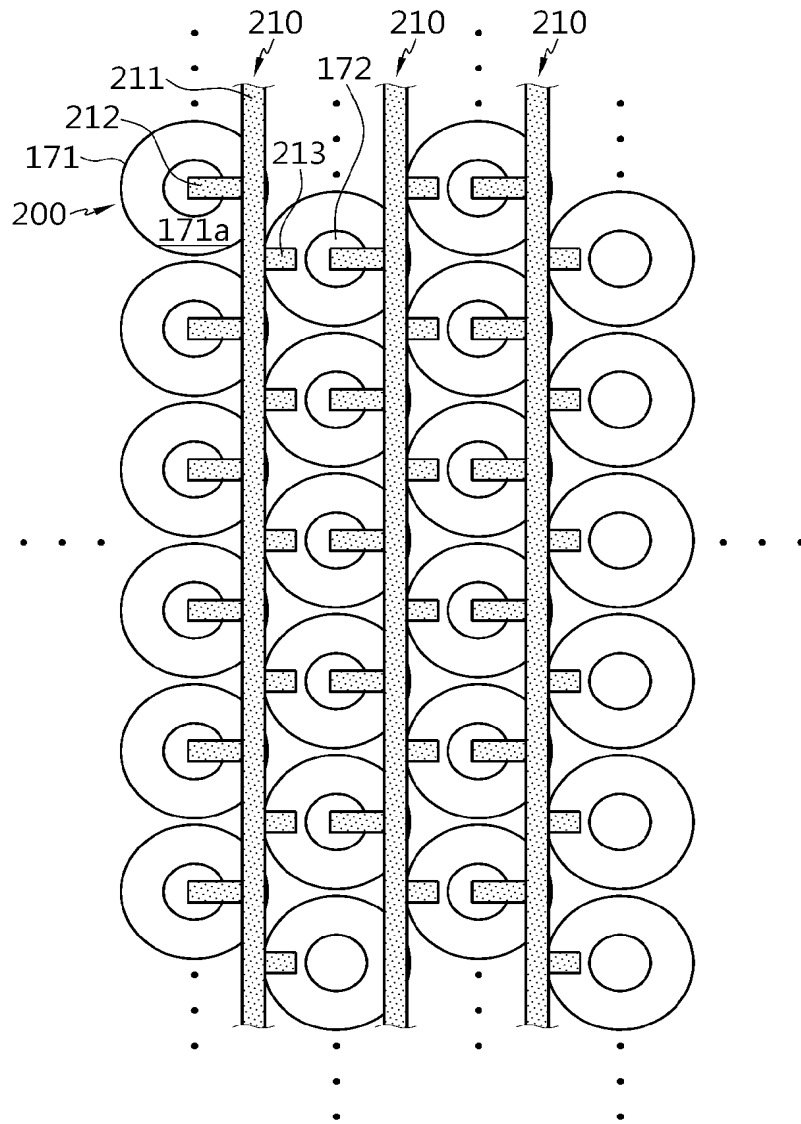
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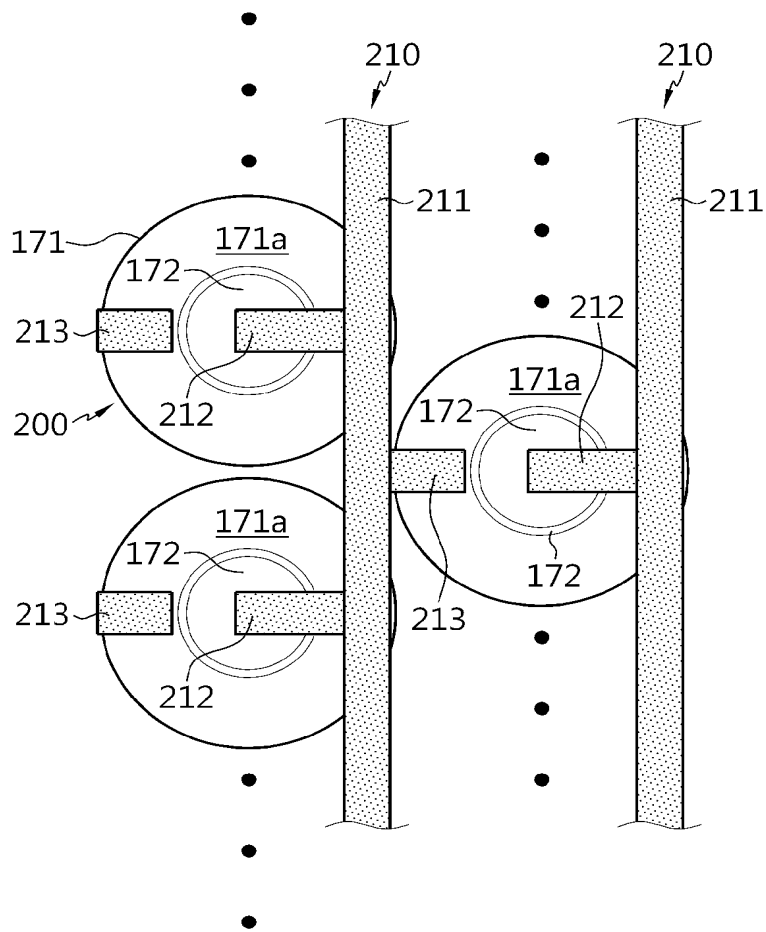
[도34]



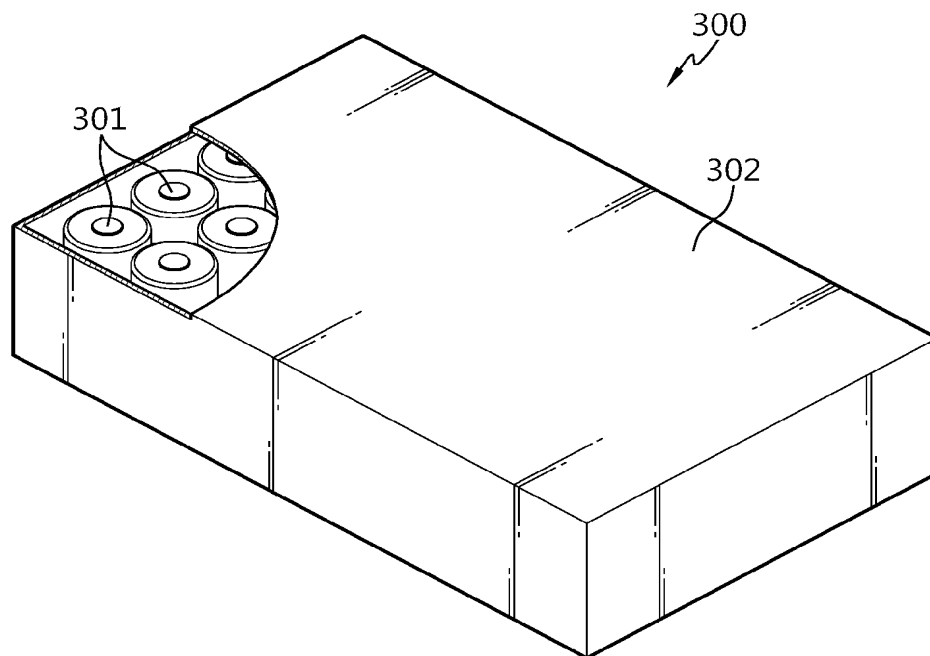
[도35]



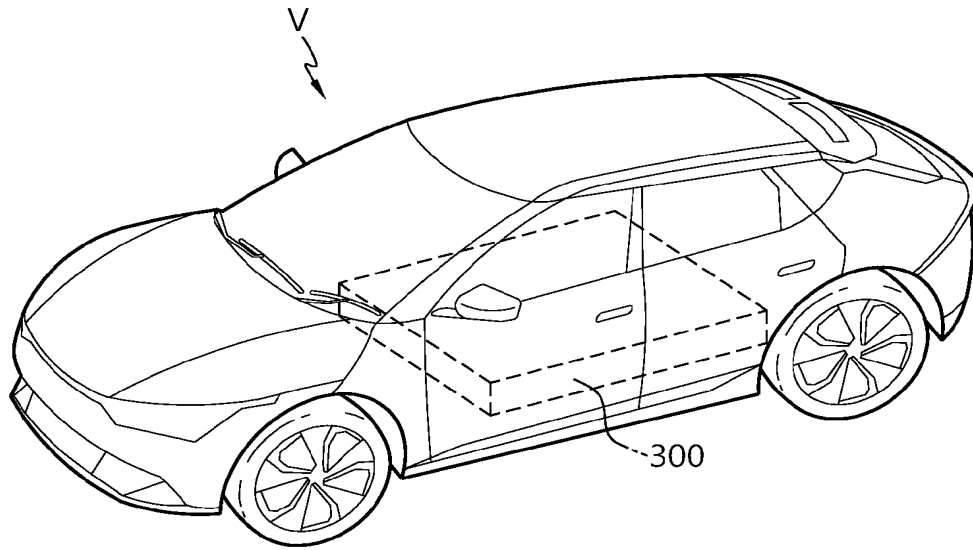
[도36]



[도37]



[도38]



INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2023/010452

A. CLASSIFICATION OF SUBJECT MATTER H01M 10/04(2006.01)i; H01M 50/531(2021.01)i; H01M 50/538(2021.01)i; H01M 50/107(2021.01)i; H01M 50/131(2021.01)i; H01M 50/213(2021.01)i According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED Minimum documentation searched (classification system followed by classification symbols) H01M 10/04(2006.01); H01M 10/052(2010.01); H01M 10/40(2006.01); H01M 2/26(2006.01); H01M 4/02(2006.01) Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched Korean utility models and applications for utility models: IPC as above Japanese utility models and applications for utility models: IPC as above Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) eKOMPASS (KIPO internal) & keywords: 원통형 배터리(cylindrical battery), 전극 조립체(electrode assembly), 권취(rolling), 응력(stress)		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 2020-084986 A1 (PANASONIC INTELLECTUAL PROPERTY MANAGEMENT CO., LTD. et al.) 30 April 2020 (2020-04-30) See paragraphs [0012]-[0023] and figures 1-2.	1-33
A	JP 2006-164773 A (MATSUSHITA ELECTRIC IND. CO., LTD.) 22 June 2006 (2006-06-22) See paragraphs [0032]-[0042] and figures 1-2.	1-33
A	JP 2008-047462 A (SONY CORP.) 28 February 2008 (2008-02-28) See paragraphs [0013]-[0048] and figures 1-9.	1-33
A	KR 10-2019-0125250 A (SUMITOMO CHEMICAL CO., LTD.) 06 November 2019 (2019-11-06) See paragraphs [0029]-[0040] and figure 1.	1-33
A	KR 10-2020-0041625 A (SAMSUNG SDI CO., LTD.) 22 April 2020 (2020-04-22) See paragraphs [0028]-[0089] and figures 1-5.	1-33
☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents: “A” document defining the general state of the art which is not considered to be of particular relevance “D” document cited by the applicant in the international application “E” earlier application or patent but published on or after the international filing date “L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) “O” document referring to an oral disclosure, use, exhibition or other means “P” document published prior to the international filing date but later than the priority date claimed “T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention “X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone “Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art “&” document member of the same patent family		
Date of the actual completion of the international search 16 November 2023		Date of mailing of the international search report 20 November 2023
Name and mailing address of the ISA/KR Korean Intellectual Property Office Government Complex-Daejeon Building 4, 189 Cheongsaro, Seo-gu, Daejeon 35208 Facsimile No. +82-42-481-8578		Authorized officer Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2023/010452

Patent document cited in search report			Publication date (day/month/year)	Patent family member(s)			Publication date (day/month/year)
WO	2020-084986	A1	30 April 2020	CN	112868126	A	28 May 2021
				EP	3872918	A1	01 September 2021
				JP	7320738	B2	04 August 2023
				US	2021-0376391	A1	02 December 2021
				WO	2020-084986	A1	30 April 2020

JP	2006-164773	A	22 June 2006	None			

JP	2008-047462	A	28 February 2008	JP	5141940	B2	13 February 2013

KR	10-2019-0125250	A	06 November 2019	CN	107204417	A	26 September 2017
				JP	2017-168327	A	21 September 2017
				JP	6240697	B2	29 November 2017
				KR	10-1890061	B1	20 August 2018
				KR	10-2017-0107916	A	26 September 2017
				KR	10-2018-0094504	A	23 August 2018
				US	2017-0271640	A1	21 September 2017
				US	2019-0312249	A1	10 October 2019

KR	10-2020-0041625	A	22 April 2020	CN	113169369	A	23 July 2021
				EP	3866241	A1	18 August 2021
				US	2021-0344033	A1	04 November 2021
				WO	2020-075990	A1	16 April 2020

A. 발명이 속하는 기술분류(국제특허분류(IPC)) H01M 10/04(2006.01)i; H01M 50/531(2021.01)i; H01M 50/538(2021.01)i; H01M 50/107(2021.01)i; H01M 50/131(2021.01)i; H01M 50/213(2021.01)i		
B. 조사된 분야 조사된 최소문헌(국제특허분류를 기재) H01M 10/04(2006.01); H01M 10/052(2010.01); H01M 10/40(2006.01); H01M 2/26(2006.01); H01M 4/02(2006.01) 조사된 기술분야에 속하는 최소문헌 이외의 문헌 한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우)) eKOMPASS(특허청 내부 검색시스템) & 키워드: 원통형 배터리(cylindrical battery), 전극 조립체(electrode assembly), 권취(rolling), 응력(stress)		
C. 관련 문헌		
카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
A	WO 2020-084986 A1 (PANASONIC INTELLECTUAL PROPERTY MANAGEMENT CO., LTD. 등) 2020.04.30 단락 [0012]-[0023] 및 도면 1-2	1-33
A	JP 2006-164773 A (MATSUSHITA ELECTRIC IND. CO., LTD.) 2006.06.22 단락 [0032]-[0042] 및 도면 1-2	1-33
A	JP 2008-047462 A (SONY CORP.) 2008.02.28 단락 [0013]-[0048] 및 도면 1-9	1-33
A	KR 10-2019-0125250 A (스미토모 가가꾸 가부시키가이샤) 2019.11.06 단락 [0029]-[0040] 및 도면 1	1-33
A	KR 10-2020-0041625 A (삼성에스디아이 주식회사) 2020.04.22 단락 [0028]-[0089] 및 도면 1-5	1-33
☐ 추가 문헌이 C(계속)에 기재되어 있습니다. ☒ 대응특허에 관한 별지를 참조하십시오.		
* 인용된 문헌의 특별 카테고리: “A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌 “D” 본 국제출원에서 출원인이 인용한 문헌 “E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후에 공개된 선출원 또는 특허 문헌 “L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌 “O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌 “P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌 “T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌 “X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다. “Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다. “&” 동일한 대응특허문헌에 속하는 문헌		
국제조사의 실제 완료일 2023년 11월 16일(16.11.2023)		국제조사보고서 발송일 2023년 11월 20일(20.11.2023)
ISA/KR의 명칭 및 우편주소 대한민국 특허청 (35208) 대전광역시 서구 청사로 189, 4동 (둔산동, 정부대전청사) 팩스 번호 +82-42-481-8578		심사관 이강하 전화번호 +82-42-481-5003

국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
WO 2020-084986 A1	2020/04/30	CN 112868126 A	2021/05/28
		EP 3872918 A1	2021/09/01
		JP 7320738 B2	2023/08/04
		US 2021-0376391 A1	2021/12/02
		WO 2020-084986 A1	2020/04/30
JP 2006-164773 A	2006/06/22	없음	
JP 2008-047462 A	2008/02/28	JP 5141940 B2	2013/02/13
KR 10-2019-0125250 A	2019/11/06	CN 107204417 A	2017/09/26
		JP 2017-168327 A	2017/09/21
		JP 6240697 B2	2017/11/29
		KR 10-1890061 B1	2018/08/20
		KR 10-2017-0107916 A	2017/09/26
		KR 10-2018-0094504 A	2018/08/23
		US 2017-0271640 A1	2017/09/21
		US 2019-0312249 A1	2019/10/10
KR 10-2020-0041625 A	2020/04/22	CN 113169369 A	2021/07/23
		EP 3866241 A1	2021/08/18
		US 2021-0344033 A1	2021/11/04
		WO 2020-075990 A1	2020/04/16



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893
86765	7590	03/25/2025	EXAMINER	
LG			DIGNAN, MICHAEL L	
Lerner David LLP			ART UNIT	PAPER NUMBER
20 Commerce Drive			1723	
Cranford, NJ 07016			NOTIFICATION DATE	DELIVERY MODE
			03/25/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com

Office Action Summary

Application No.

18/848,306

Applicant(s)

Kwon et al.

Examiner

MICHAEL L DIGNAN

Art Unit

1723

AIA (FITF) Status

Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 1/8/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-33 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1-33 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 1/8/025 is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☒ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

DETAILED ACTION

Notice to Applicant

1. Claims 1-33 are pending and are examined herein. This is the first action on the merits.

Claim Rejections - 35 USC § 112

2. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL. —The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

3. **Claims 19 and 25 rejected under 35 U.S.C. 112(a) as failing to comply with the written description** requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

Claim 19 has been amended to require “the core side end of the second electrode extends further in a direction opposite to the **winding axis direction**” as well as an outer circumference side end referring to a “winding axis direction.” Support for this limitation might appear to come from paragraph 0173 of the as-filed specification, which does not refer to a “winding axis direction” but a “winding direction,” which appear to be substantively different. There is no clear support for a “winding axis direction” extension difference as claimed.

Claim 25 suffers from a similar problem, where the amended “winding axis direction” appears to subvert the expectation that the segments are actually placed along a winding direction as described in the specification.

Art Unit: 1723

4. The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.
5. **Claims 1-33 are rejected under 35 U.S.C. 112(b) as being indefinite** for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor regards as the invention.

Claim 1 says “**an outer circumference of the first fan-shaped region is defined as a stress vulnerable region**” and “**an outer circumference of the second fan-shaped region is defined as a stress amplification region.**” It is unclear whether the “a stress vulnerable region” and the “a stress amplification region” are just arbitrary names for arcs along the outer circumference of the jelly roll or are intended to reflect non-arbitrary structural features of the defined regions. If the latter, it is further unclear what the metes and bounds of those structural features are. The as-filed specification says that these regions can “change their angles and positions” upon cycling (para 0186) but it is unclear whether this is inherent to all jelly rolls or only to jelly rolls with a particular composition or structure. If the latter, it is unclear what that particular composition/structure is. For purposes of examination, the Office has interpreted the limitations broadly to be arbitrary names for a designated arc section of the outer circumference. The dependent claims are rejected for depending on claim 1.

Claim 19 says “in a direction opposite to the winding axis direction.” It is unclear what this means, since the specification appears to indicate a longer second electrode than the first electrode in the winding direction, not a taller electrode. The claim has been interpreted in line with the specification to refer a second electrode with longer inner and outer winding directions. Claim 25 is indefinite because it is unclear how to interpret “winding axis direction.” It has been interpreted as a winding direction along the electrode.

Double Patenting

6. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration

while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

7. **Claims 1-33 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 3, and 5-20 of copending Application No. 18/670,134**, in which the Office has issued a Notice of Allowance on 1/6/2025, but which has not proceeded to issue as a patent of the date of the instant office action. Although the claims at issue are not identical, they are not patentably distinct from each other.

The '134 application is directed towards an electrode assembly with a spirally wound jelly roll in a casing, wherein two arcs are defined by the inner/outer ends of the first/second electrodes, as in instant claim 1, and wherein they are non-overlapping such that a bisecting diameter can be drawn and the two arcs would be in separate semicircles (see also instant claim 4, claiming essentially the same thing). The instant '134 application does not refer to "stress amplification" or "stress vulnerable" regions, but as pointed above, it is unclear whether this is intended to be a positive structural feature (or what that feature might be) or just an arbitrary name for the arcs as defined.

The '134 application is also directed towards third arc sectors (claim 5), arbitrary angles capable of bisection (claims 6-8), sub-90° arcs (claim 12), winding turns of the separator (claim 16), and otherwise conventional features of cylindrical cells, such as a casing connected to one electrode, a plug

for the casing, etc. Any instant claims not found directly in the '134 application appear to involve geometric limitations that are either intrinsically part of the jelly roll of the '134 application or are selected from a relatively small finite set of geometric possibilities, in combination with conventional jelly roll features. A structure or method step that is obvious to try— such as one that is chosen from a finite number of identified, predictable solutions, with a reasonable expectation of success, has been found to be obvious. See *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007).

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claim Rejections - 35 USC § 102

8. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

9. **Claims 1-3, 13-19 and 22-23 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Okuda (US 2021/0376391 to Okuda et al.).**

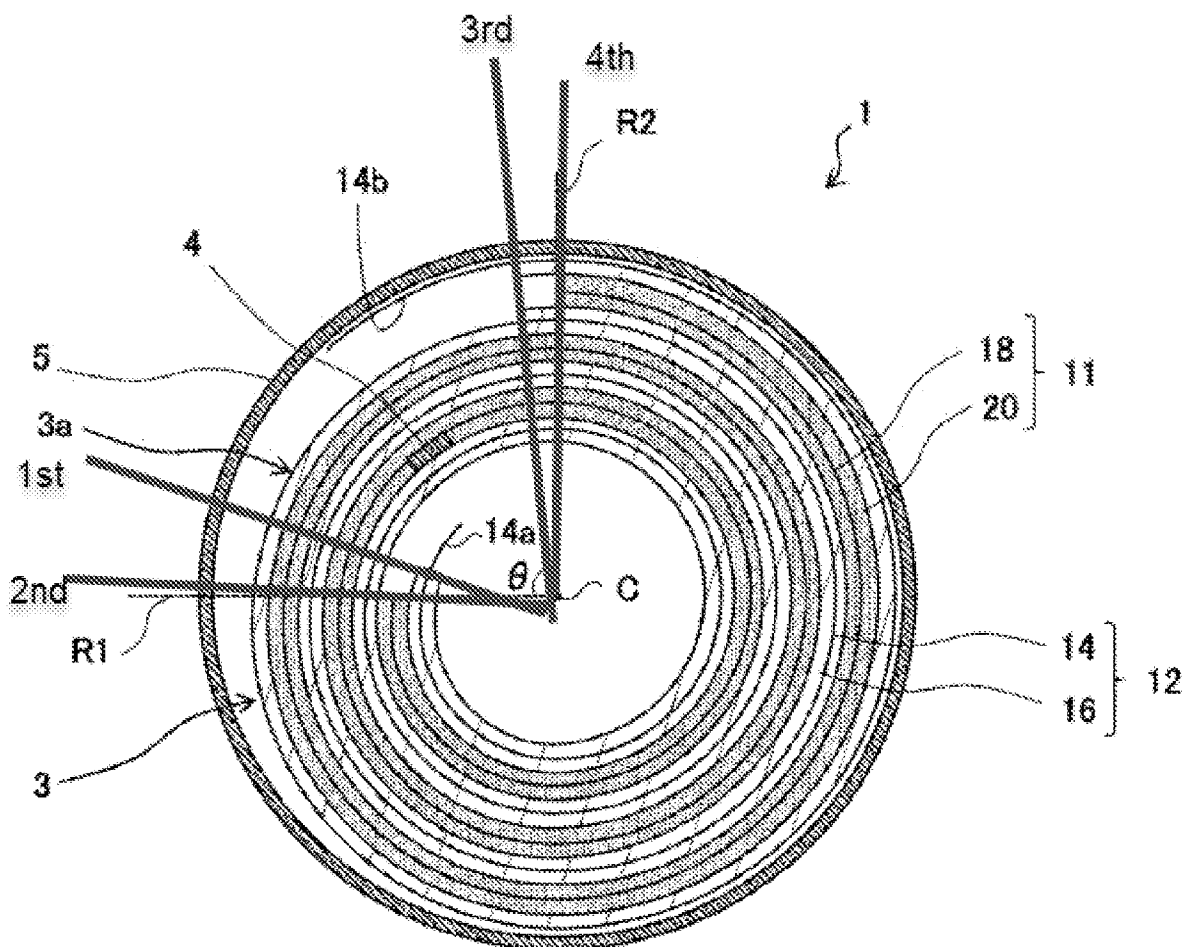
Regarding Claim 1, Okuda teaches:

- a cylindrical battery (abstract) comprising a wound electrode assembly with cathode, anode, and separator around a winding axis to define a core and an outer circumference (Fig. 2)
- a battery housing 5 accommodating the assembly (para 0015, Fig. 2)

a first fan-shaped region defined by a line passing from the center through the core side end of the first electrode and a second line passing from the center through the core side end of the second electrode (Fig. 2)

- a second fan-shaped region defined by a third line passing through a center and an outer end of the first electrode and a fourth line passing through a center and an outer end of the second electrode (Fig. 2)
- the electrode assembly having a winding structure with an outer circumference side end of the first electrode in the second fan region spaced apart from an inside of the first fan region along a circumference (Fig. 2)

Figure 2



Regarding Claim 2, Okuda teaches:

- wherein an outer end of the first electrode **14** is in contact with an inner surface of the battery housing (Fig. 2, para 0019)

Regarding Claim 3, Okuda teaches:

- wherein the two fan regions are spaced apart along the circumferential direction of the cross section (Fig. 2)

Regarding Claim 13, Okuda teaches:

under an interpretation with first and second electrode designations swapped from the above annotated Fig. 2 (i.e. where 1st/2nd and 3rd/4th lines are swapped), wherein the angle between swapped 1st and 4th lines is greater than that between the swapped 2nd and 3rd lines

Regarding Claim 14, Okuda teaches:

- an angle between 2nd and 4th straight lines greater than 1st and 3rd (Fig. 2)

Regarding Claim 15, Okuda teaches:

- second fan angle smaller than first fan angle (Fig. 2)

Regarding Claim 16, Okuda teaches:

- capacity retention through 300 cycles, indicating a “winding structure” that is maintained within the broadest reasonable interpretation of the phrase (Table 1)

Regarding Claim 17, Okuda teaches:

- a plurality of winding turns for the electrodes, with arbitrarily selectable “first” and “second” electrodes for polarity, wherein at least a part of the second negative electrode turns are near a core side and a part extend radially (Fig. 2)

Regarding Claim 18, Okuda teaches:

- wherein the separator also includes a plurality of winding turns provided at an inner side (Fig. 2)

Regarding Claim 19, Okuda teaches:

- a longer negative electrode in the winding direction both in the core and at the outer circumference (Fig. 2)

Regarding Claim 22, Okuda teaches:

- an open end and a bottom portion in the housing, the housing connected to the first electrode (Fig. 1, para 0002, etc.)

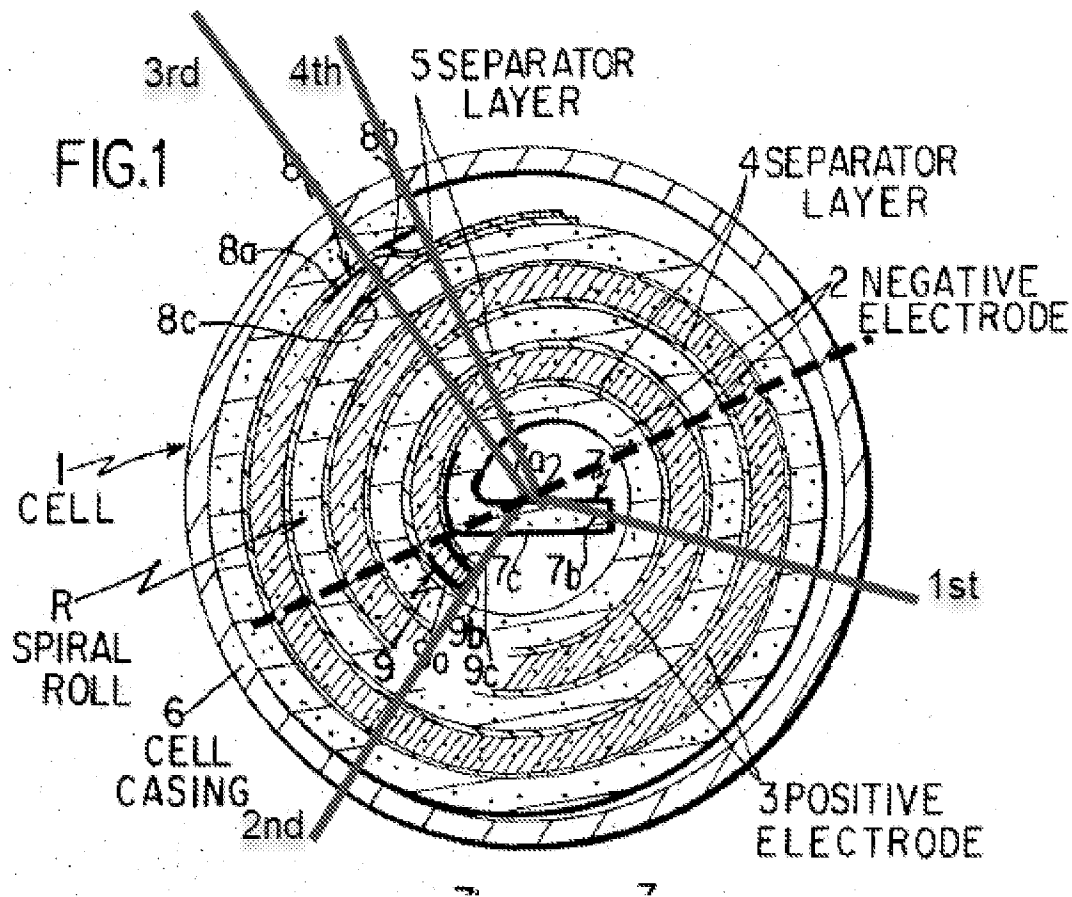
Regarding Claim 23, Okuda teaches:

- a sealing body for sealing the open end including a terminal for connecting to the non-housing electrode exposed to the exterior (Fig. 1)

10. Claims 1-6, 9-15, 17-23, and 30 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Binder (US Patent No. 3,298,871 to Binder et al.)

Regarding Claim 1, Binder teaches:

- a cylindrical battery comprising an electrode assembly of first electrode, second electrode, and separator wound around a winding axis to define a core and an outer circumference (Fig. 1)



- a housing to accommodate the assembly (Fig. 1, column 4)
- a first fan-shaped region defined by radial lines extending from the center through the core ends of the electrodes and a second fan-shaped region defined by radial lines extend from the center through the outer ends of the electrodes (Fig. 1)
- the outer end of the first electrode is spaced apart from an inside of first fan-shaped region (Fig. 1)

Regarding Claim 2, Binder teaches:

- wherein the outer end of the first electrode contact an inner surface of the battery housing (Fig. 1)

Regarding Claim 3, Binder teaches:

- wherein the second fan region is spaced apart from the first fan region along the circumferential direction (Fig. 1)

Regarding Claim 4, Binder teaches:

- wherein the fan-shaped regions are in opposite semicircles defined by a diameter perpendicular to a radial line bisecting the first fan-shaped region (Fig. 1)

Regarding Claim 5, Binder teaches:

- wherein at least part of the first fan region overlaps with a third fan region that is point symmetric to the second fan region relative to the center (Fig. 1)

Regarding Claim 6, Binder teaches:

- wherein at least part of the first fan region overlaps with a fifth straight line bisecting the third point symmetric fan region (Fig. 1)

Regarding Claims 9-12, Binder teaches:

- wherein when a fifth straight line that bisects a fan region point symmetric with the first fan region is drawn, the fourth straight line is located between the fifth straight line and the first line straight line and the third straight line is located between the fourth straight line and the second straight line, or vice versa, either in the large angle or the small formed (Fig. 2)

Regarding Claims 13, Binder teaches:

- an angle between 1st and 4th greater than that between 2nd and 3rd (Fig. 2)

Regarding Claim 14, Binder teaches:

- an angle between 2nd and 4th greater than an angle between 1st and 3rd, if the first electrode and second electrode designations adopted above are swapped (Fig. 2, swapped)

Regarding Claim 15, Binder teaches:

- a second fan region angle smaller than first fan region angle (Fig. 2)

Regarding Claim 17, Binder teaches:

- wherein the first electrode can be the positive one, and the second negative, wherein both include a plurality of winding turns, including some near the core and some extending in a radial direction (Fig. 1)

Regarding Claim 18, Binder teaches:

- separator with a plurality of winding turns at an inner side (Fig.1)

Regarding Claim 19, Binder teaches:

- a negative electrode longer in the winding direction both in the core and at the outer circumference (Fig. 1)

Regarding Claim 20, Binder teaches:

- a first fan region between 30° and 180° and a second fan region between 10° and 90° (Fig. 1)

Regarding Claim 21, Binder teaches:

- a first fan region of more than 90° and a second fan region of less than 30°

Regarding Claims 22-23, Binder teaches:

- a housing with an open end covered by a sealing terminal with an exterior portion, a bottom, and wherein the housing is one of the terminals (column 4, Figs.)

Regarding Claim 30, Binder teaches:

- a diameter to a height greater than 0.4 (Fig. 1A)

Claim Rejections - 35 USC § 103

11. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

12. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or

nonobviousness.

13. Claims 7-8 and 31-33 are rejected under 35 U.S.C. 103 as being unpatentable over Binder (US Patent No. 3,298,871 to Binder et al.).

Regarding Claims 7-8, Binder does not explicitly teach:

- the particular position of the second fan region relative to a third region point symmetric to the first region

Binder, however, does not particularly describe the relative position of the ends along the circumference, other than to position it in the opposite semicircle from the first fan region. A structure or method step that is obvious to try— such as one that is chosen from a finite number of identified, predictable solutions, with a reasonable expectation of success, has been found to be obvious. See *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007). It would have been obvious to shift the second fan region clockwise/counterclockwise slightly to read on claims 7-8, since such positions appear to be functionally equivalent, depending only on the length of the electrodes, and therefore subject to change depending on particular desired capacity given a set of electrodes with defined thickness and other properties, at least absent evidence of unexpected results for the particularly claimed radial position of the second fan region relative to what is shown in Binder or otherwise rendered obvious by the prior art.

Regarding Claim 31, Binder does not teach:

- the claimed form factor

Those form factors are well-known commercial standards that would be obvious. Where the only difference between the prior art and the claims is a recitation of relative dimensions of the claimed device and a device having the claimed relative dimensions would not perform differently than the prior art device the claimed device is not patentably distinct from the prior art device. See *Gardner v. TEC Systems, Inc.*, 725 F.2d 1338, 220 USPQ 777 (Fed. Cir. 1984); and MPEP 2144 IV. A.

Regarding Claims 32 and 33, Binder does not teach:

- a plurality of batteries
- a vehicle comprising the plurality of batteries

Assembling battery cells into packs was well-known, as was using battery packs in vehicles. Use of a known technique to improve similar devices, methods, or products in the same way, and applying a known technique to a known device, method, or product ready for improvement to yield predictable results has been found to be obvious. See *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007).

14. Claims 24-29 are rejected under 35 U.S.C. 103 as being unpatentable over Okuda (US 2021/0376391 to Okuda et al.) in view of Park (US 2021/0344033 to Park et al.)

Regarding Claims 24-29, Okuda teaches:

- positive and negative electrodes

Okuda does not explicitly teach:

- where each of the electrodes include an uncoated portion at a long side extending along the winding axis direction of the electrode assembly
- the uncoated portions protruding outwards from the separator and in a bent state extending in a radial direction to form a bending surface coupled to respective current collectors
- the uncoated regions with a plurality of segments along the winding direction
- the coupling regions having a radial structure

Park, however, from the same field of invention, teaches an uncoated portion extending in opposite directions for respective electrodes, and having segments (Fig. 2), the segments bent to connect to respective current collectors forming a radial coupling region (Fig. 1). Use of a known technique to improve similar devices, methods, or products in the same way, and applying a known technique to a known device, method, or product ready for improvement to yield predictable results has been found to be obvious. See *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007). In the instant case, it would have been obvious to provide the uncoated portions and current collectors as claimed, since this was a conventional terminal-forming method in the art, as taught in Park.

Conclusion

15. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure:

- US Patent No. 5,508,122
- US Patent No. 6,391,488
- US 2005/0031950
- US 2016/0218368
- US 2022/0123373

16. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Michael Dignan, whose telephone number is (571) 272-6425. The examiner can normally be reached from Monday to Friday between 10 AM and 6:30 PM. If any attempt to reach the examiner by telephone is unsuccessful, the examiner's supervisor, Tiffany Legette, can be reached at (571)270-7078. Another resource that is available to applicants is the Patent Application Information Retrieval (PAIR). Information regarding the status of an application can be obtained from the (PAIR) system. Status information for published applications may be obtained from either Private PAIR or

Art Unit: 1723

Public PAX. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, please feel free to contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). Applicants are invited to contact the Office to schedule an in-person interview to discuss and resolve the issues set forth in this Office Action. Although an interview is not required, the Office believes that an interview can be of use to resolve any issues related to a patent application in an efficient and prompt manner.

/MICHAEL L DIGNAN/
Examiner, Art Unit 1723

<i>Notice of References Cited</i>	Application/Control No. 18/848,306		Applicant(s)/Patent Under Reexamination Kwon et al.	
	Examiner MICHAEL L DIGNAN		Art Unit 1723	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-3298871-A	1967-01-17	FELIX LECOUFFE YVES JACQUES	H01M6/10	429/94
*	B	US-5508122-A	1996-04-16	Narukawa; Satoshi	H01M6/10	429/94
*	C	US-6391488-B1	2002-05-21	Shimizu; Toshiyuki	H01M6/10	429/231.95
*	D	US-20050031950-A1	2005-02-10	Maeda, Taishi	H01M10/345	429/218.2
*	E	US-20160218368-A1	2016-07-28	YAMADA; Ichiro	H02J7/007194	1/1
*	F	US-20210344033-A1	2021-11-04	PARK; Jung Hyun	H01M50/152	1/1
*	G	US-20210376391-A1	2021-12-02	Okuda; Hiroyuki	H01M10/0431	1/1
*	H	US-20220123373-A1	2022-04-21	TAKAHASHI; Masashi	B64D27/34	1/1
	I					
	J					
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	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 18/848,306	Applicant(s)/Patent Under Reexamination Kwon et al.
	Examiner MICHAEL L DIGNAN	Art Unit 1723

CPC - Searched*		
Symbol	Date	Examiner
H01M 10/0431, 2220/20, 10/0422	03/20/2025	Michael Dignan

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
performed inventor/assignee searches, reviewed IDS and search reports, reviewed global dossier, performed CPC search with text in PE2E	03/20/2025	Michael Dignan

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/MICHAEL L DIGNAN/ Examiner, Art Unit 1723	
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<i>Index of Claims</i> 	Application/Control No. 18/848,306	Applicant(s)/Patent Under Reexamination Kwon et al.
	Examiner MICHAEL L DIGNAN	Art Unit 1723

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	03/20/2025								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
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	29	✓								
	30	✓								
	31	✓								
	32	✓								
	33	✓								

Bibliographic Data

Application No:18/848,306

Foreign Priority claimed:

☒ Yes

☐ No

35 USC 119 (a-d) conditions met:

☒ Yes

☐ No

☐ Met After Allowance

Verified and Acknowledged:

/MICHAEL L DIGNAN/

Examiner's Signature

Initials

Title:

Cylindrical Battery, Battery Pack and Vehicle

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/08/2025	429	1723	
RULE			

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Ju-Nam Son, Daejeon, KOREA, REPUBLIC OF
Seo-Hyeon Chae, Daejeon, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a 371 of PCT/KR2023/010452 07/19/2023

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF KR10-2022-0089226 07/19/2022
KOREA, REPUBLIC OF KR10-2023-0094183 07/19/2023

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Doc description: Information Disclosure Statement (IDS) Filed

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18848306	
	Filing Date		2024-09-18	
	First Named Inventor	Yong-Hwan Kwon		
	Art Unit		N/A	
	Examiner Name	Not Yet Assigned		
	Attorney Docket Number		LGES 3.3F-5465	

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	
	1	6677082	B2	2004-01-13	Thackeray et al.		
	2	6680143	B2	2004-01-20	Thackeray et al.		
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	1	20070196730	A1	2007-08-23	Kozuki, Kiyomi		
	2	20160351966	A1	2016-12-01	Batson et al.		
	3	20170271640	A1	2017-09-21	SAKAMOTO, Akinobu		
	4	20210344033	A1	2021-11-04	PARK et al.		

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Art Unit	N/A
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5	20210376391	A1	2021-12-02	Okuda et al.	
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	1	205319258	CN	U	2016-06-15	Chen, Pei	Machine generated translation is attached.	☒
	2	2006164773	JP	A	2006-06-22	Matsushita Electric Ind Co Ltd	Machine generated translation is attached.	☒
	3	2008047462	JP	A	2008-02-28	Sony Corp	Machine generated translation is attached.	☒
	4	5082256	JP	B2	2012-11-28	Matsushita Electric Ind Co Ltd	U.S. Publication No. 20070196730 A1 is an English language counterpart.	☒
	5	102262668	KR	B1	2021-06-09	A123 SYSTEMS LLC	U.S. Publication No. 20160351966 A1 is an English language counterpart.	☒
	6	20190125250	KR	A	2019-11-06	Sumitomo Chemical Co., Ltd	U.S. Publication No. 20170271640 A1 is an English language counterpart.	☒
	7	20200041625	KR	A	2020-04-22	Samsung Sdi Co Ltd	U.S. Publication No. 20210344033 A1 is an English language counterpart.	☒
	8	2020084986	WO	A1	2020-04-30	Panasonic Ip Man Co Ltd	U.S. Publication No. 20210376391 A1 is an English language counterpart.	☒

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	Art Unit		N/A
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	Attorney Docket Number		LGES 3.3F-5465

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	
	1	International Search Report for Application Number PCT/KR2023/010452 mailed November 20, 2023, pages 1-3.	☒	
	2	Written Opinion of The International Searching Authority, for Application Number PCT/KR2023/010452, dated 11/20/2023, pages 1 - 5.	☒	
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Examiner Signature	/MICHAEL L DIGNAN/		Date Considered	03/19/2025
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CERTIFICATION STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

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That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

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That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

- ☐ See attached certification statement.
- ☒ A certification statement is not submitted herewith.

FEE

- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

COPIES

- ☐ An identification of an earlier application pursuant to 37 CFR 1.98 (d) (1) is attached.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Matthew J. Shatynski/	Date (YYYY-MM-DD)	2024-09-19
Name/Print	Matthew J. Shatynski	Registration Number	81,269

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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	First Named Inventor	Yong-Hwan Kwon		
	Art Unit	N/A		
	Examiner Name	Not Yet Assigned		
	Attorney Docket Number	LGES 3.3F-5465		

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	1	20050031950	A1	2005-02-10	Maeda, Taishi			
	2	20160218368	A1	2016-07-28	YAMADA et al.			
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First Named Inventor		Yong-Hwan Kwon	
Art Unit		N/A	
Examiner Name		Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-5465	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
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- ☐ See attached certification statement.
- ☒ A certification statement is not submitted herewith.

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Signature	/Matthew J. Shatynski/	Date (YYYY-MM-DD)	2024-09-27
Name/Print	Matthew J. Shatynski	Registration Number	81,269

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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	0	"18848306".apnr.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/17 01:25 PM
L2	260921	(H01M10/0431 OR H01M2220/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/17 01:31 PM
L3	50	(Artificial Intelligence) Similar to: 18/848,306 with 0 CPC Selections and 0 Text Selections Text: (EP-4513664-A1 OR CN-115312877-A OR KR-20230028162-A OR EP-2819204-B1 OR JP-5191993-B2 OR EP-2337120-A2 OR EP-4468495-A1 OR CN-111837255-B OR US-9515309-B2 OR US-20230087166-A1 OR KR-100615153-B1 OR WO-2021006529-A1 OR US-11038218-B2 OR CN-115064757-B	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2025/03/19 02:51 PM

		OR CN-118975031-A OR US-20100143787-A1 OR DE-102018204165-A1 OR CN-222088800-U OR WO-2008056908-A1 OR KR-20240169982-A OR US-20240356126-A1 OR CN-116420273-A OR CN-118117250-A OR CN-119096403-A OR US-20250070218-A1 OR EP-4394996-A1 OR CN-105374956-A OR US-8114536-B2 OR KR-20210006712-A OR CN-102005602-A OR US-20230079811-A1 OR US-9070929-B2 OR EP-2772960-A1 OR CN-222126934-U OR CN-118610669-A OR WO-2024021025-A1 OR EP-4297144-A1 OR CN-219959089-U OR US-20190131594-A1 OR KR-20150055775-A OR KR-100509747-B1 OR CN-116783723-A OR EP-2772960-B1 OR US-20140050960-A1 OR KR-20240170003-A OR KR-20190046404-A OR WO-2018071825-A3 OR KR-101653305-B1 OR EP-4443632-A1 OR CN-114069167-B).did.					
L4	9	("20050031950" OR "20160218368" OR "20070196730" OR "20160351966" OR "20170271640" OR "20210344033" OR "20210376391" OR "6677082" OR "6680143").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/19 02:54 PM
L5	267177	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL,	OR	ON	ON	2025/03/20 10:40 AM

L6	13474	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc. AND jelly ADJ roll	PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/20 10:40 AM
L7	2522	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc. AND jelly ADJ roll	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 10:40 AM
L8	1	(17/282094).APP.	(USPAT; USOCR)	OR	ON	ON	2025/03/20 10:51 AM
L9	8	("20030186095" "20130266834" "20140045036" "20140141303" "20150353396" "20160218368" "20180040881" "20210119263").pn. OR ("12155041").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 10:51 AM
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L11	860	L2 AND (roll\$3 wind\$3 wound) SAME (arc fan) AND (end)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:16 AM
L12	869	L5 AND (roll\$3 wind\$3 wound) SAME (arc fan) AND (end)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:45 AM
L13	3	L5 AND (roll\$3 wind\$3 wound) SAME (arc fan) AND (end) AND ("500" ADJ cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:45 AM
L14	4	L5 AND ("500" ADJ	(US-PGPUB; USPAT;	OR	ON	ON	2025/03/20

		cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	USOCR)				11:49 AM
L15	17	L5 AND (retention SAME cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:50 AM
L16	51	h01m AND (retention SAME cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:52 AM
L17	166	h01m AND (cell form) WITH ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:54 AM
L18	3356	L5 AND (cycles) SAME ("200" "300" "400" "500" "600" "700" "800" "900")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:56 AM
L19	1299	L5 AND (cycles) SAME ("200" "300" "400" "500" "600" "700" "800" "900") AND cylindrical AND (wound wind\$3 roll\$3)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:57 AM
L20	3	"5965290".pn. "6391488".pn. "20110014509".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L21	0	(12/835020).APP.	(USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L22	7	("2011/0014509").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L23	21	("4963445" "5965290").pn. OR ("6391488").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L24	18	("3298871" "4707421" "5354629" "5458993" "5508122").pn. OR ("5965290").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L25	42	L22 L23 L24	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L26	2522	L5 AND (jelly ADJ roll)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:05 PM
L27	12284	L5 AND (jelly ADJ roll)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA,	OR	ON	ON	2025/03/20 12:05 PM

			EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO))				
L28	8780	L5 AND (jelly ADJ roll) AND cylindrical	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO))	OR	ON	ON	2025/03/20 12:05 PM
L29	1684	L5 AND (jelly ADJ roll) AND cylindrical	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:05 PM
L30	396	L5 AND (jelly ADJ roll) AND cylindrical AND core	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:05 PM

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	33686	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:52 AM
N2	2	N1 AND cylindrical.clm. AND (fan ADJ shaped).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:52 AM
N3	0	N1 AND cylindrical.clm. AND (arc).clm. AND stress\$3.clm. AND (wound roll).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:53 AM
N4	0	N1 AND cylindrical.clm. AND (arc).clm. AND stress\$3.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:53 AM

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Application	Document	Mailroom Date	Attorney Docket No.
18848306	CTNF	03/25/2025	LGES 3.11F-5465
18848306	892	03/25/2025	LGES 3.11F-5465
18848306	1449	03/25/2025	LGES 3.11F-5465
18848306	1449	03/25/2025	LGES 3.11F-5465

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465

CONFIRMATION NO. 7893

86765
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

PUBLICATION NOTICE



OC00000089092604

Date Mailed: 05/22/2025

Title:Cylindrical Battery, Battery Pack and Vehicle

Publication No.US-2025-0167278-A1

Publication Date:05/22/2025

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be viewed using the USPTO's publicly available Searchable Databases via the Patent Public Search tool at www.uspto.gov. The direct link to access the Patent Public Search tool is currently <https://ppubs.uspto.gov/pubwebapp/static/pages/ppubsbasic.html>.

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18848306	NTC.PUB	05/22/2025	LGES 3.11F-5465

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893
86765	7590	06/02/2025	EXAMINER	
LG			DIGNAN, MICHAEL L	
Lerner David LLP			ART UNIT	PAPER NUMBER
20 Commerce Drive			1723	
Cranford, NJ 07016			NOTIFICATION DATE	DELIVERY MODE
			06/02/2025	ELECTRONIC

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The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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<i>Applicant-Initiated Interview Summary</i>	Application No. 18/848,306	Applicant(s) Kwon et al.		
	Examiner MICHAEL L DIGNAN	Art Unit 1723	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
MICHAEL L DIGNAN	Primary Examiner	Telephonic
Matthew Shatynski	Attorney	

Date of Interview: 29 May 2025

Issues Discussed:

Proposed Amendment(s)

Amendments were discussed to overcome the 112 rejections and 103 rejections. No explicit agreement was reached.

/MICHAEL L DIGNAN/ Examiner, Art Unit 1723	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

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Application	Document	Mailroom Date	Attorney Docket No.
18848306	EXIN	06/02/2025	LGES 3.11F-5465

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APPLICATION #
18/848,306

RECEIPT DATE / TIME
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ATTORNEY DOCKET #
LGES 3.11F-5465

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	7893	FILED BY	Sharon McGee
PATENT CENTER #	71076398	FILING DATE	01/08/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Yong-Hwan Kwon
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Kevin Kocun

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
LGES5310 - Amendment LD-447.pdf	13	Amendment/Request for Reconsideration-After Non-Final Rejection	75 KB
LGES5310 - Transmittal -- Amendment.pdf	1	Miscellaneous Incoming Letter	21 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
LGES5310 - Amendment LD-447.pdf	DE5BAC4461B82071FC6A0EDB2367F4A1A4A1EDF0DAB119E00049D5CCA907101A6AFE4EE697D1789105830F120AF2E93CA

BC40C5B21AD2817A7D250E500C21B91

LGES5310 - Transmittal --
Amendment.pdf

A5B11292B64EB32812F57F07C1533C3F65EBD8BA9991D7A55
CAE810E2E629F96A1DD19A1DBDF4333C94B23AE233C3180C
1ACD3C03104A025FCB0E932405D9FEC

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New International Application Filed with the USPTO as a Receiving Office

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Dated: June 25, 2025

Electronic Signature for Matthew J. Shatynski: /Matthew J. Shatynski/

Docket No. LGES 3.11F-5465
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kwon et al.

Application No.: 18/848,306

Confirmation No.: 7893

Filed: January 8, 2025

Art Unit: 1723

For: Cylindrical Battery, Battery Pack and Vehicle

Examiner: M. L. Dignan

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Dear Commissioner:

In response to the official action mailed March 25, 2025, applicant submits the following amendments and remarks.

IN THE CLAIMS

1. (currently amended) A cylindrical battery, comprising:
an electrode assembly in which a first electrode, a second electrode, and a separator interposed therebetween are wound around a winding axis of the electrode assembly, so as to define a hollow core and an outer circumference of the electrode assembly; and
a cylindrical battery housing accommodating the electrode assembly,
wherein a cross-section of the electrode assembly perpendicular to a winding axis direction includes a first fan-shaped region and a second fan-shaped region,
wherein the first fan-shaped region is defined by a first straight line and a second straight line passing from a center of the core through a core side end of the first electrode and a core side end of the second electrode, respectively, and an outer circumference of the first fan-shaped region is defined as a stress vulnerable region, and
wherein the second fan-shaped region is defined by a third straight line and a fourth straight line passing from the center of the core through an outer circumference side end of the first electrode and an outer circumference side end of the second electrode, respectively, and an outer circumference of the second fan-shaped region is defined as a stress amplification region, and
wherein the electrode assembly has a winding structure in which at least the outer circumference side end of the first electrode of the stress amplification region is spaced apart from an inside of the stress vulnerable region along a circumferential direction of the cross-section of the electrode assembly,
wherein in the winding structure, a circumferential angle of the stress vulnerable region has an angle of 180 degrees or less, and a circumferential angle of the stress amplification region has an angle in a range of 10 degrees to 90 degrees.

2. (previously presented) The cylindrical battery according to claim 1,
wherein a region of the outer circumference of the electrode assembly including the outer circumference side end of the first electrode contacts an inner surface of the battery housing.

3. (previously presented) The cylindrical battery according to claim 1,
wherein in the winding structure, the stress amplification region is spaced apart from the

stress vulnerable region along the circumferential direction of the cross-section of the electrode assembly.

4. (previously presented)The cylindrical battery according to claim 1,
wherein the cross-section includes a first semicircular region and a second semicircular region based on a diametric line segment,
wherein the diametric line segment passes through a center of the core and is perpendicular to a straight line dividing a circumferential angle of the stress vulnerable region into equal angles,
wherein in the winding structure, the stress vulnerable region is located in the first semicircular region and the stress amplification region is located in the second semicircular region.

5. (previously presented)The cylindrical battery according to claim 1,
wherein in the winding structure at least a part of the stress amplification region overlaps with a third fan-shaped region in a second semicircular region, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core.

6. (previously presented)The cylindrical battery according to claim 5,
wherein in the winding structure at least a part of the stress amplification region overlaps with a fifth straight line that divides a circumferential angle of the third fan-shaped region into equal angles.

7. (previously presented)The cylindrical battery according to claim 1,
wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and
wherein in the winding structure the third straight line and the fourth straight line are located between the second straight line and the fifth straight line based on the circumferential direction of the cross-section.

8. (previously presented)The cylindrical battery according to claim 1,
wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein

the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and

wherein in the winding structure the third straight line and the fourth straight line are located between the first straight line and the fifth straight line based on the circumferential direction on the cross-section.

9. (previously presented)The cylindrical battery according to claim 1,

wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and

wherein in the winding structure the fourth straight line is located between the fifth straight line and the first straight line, and the third straight line is located between the fourth straight line and the second straight line based on the circumferential direction on the cross-section.

10. (previously presented)The cylindrical battery according to claim 1,

wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and

wherein in the winding structure the fourth straight line is located between the fifth straight line and the first straight line, and the third straight line is located between the fifth straight line and the second straight line based on the circumferential direction on the cross-section.

11. (previously presented)The cylindrical battery according to claim 1,

wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and

wherein in the winding structure the fourth straight line is located between the fifth straight line and the second straight line, and the third straight line is located between the fourth straight line and the first straight line based on the circumferential direction on the cross-section.

12. (previously presented)The cylindrical battery according to claim 1,
wherein when a fifth straight line divides a third fan-shaped region into equal angles, wherein the third fan-shaped region is point symmetrical with the stress vulnerable region based on the center of the core, and

wherein in the winding structure the fourth straight line is located between the fifth straight line and the second straight line, and the third straight line is located between the fifth straight line and the first straight line based on the circumferential direction on the cross-section.

13. (previously presented)The cylindrical battery according to claim 1,
wherein the winding structure has a circumferential angle between the first straight line and the fourth straight line that is greater than a circumferential angle between the second straight line and the third straight line, based on the circumferential direction on the cross-section.

14. (previously presented)The cylindrical battery according to claim 1,
wherein the winding structure has a circumferential angle between the second straight line and the fourth straight line that is greater than a circumferential angle between the first straight line and the third straight line, based on the circumferential direction on the cross-section.

15. (previously presented)The cylindrical battery according to claim 1,
wherein the winding structure has a circumferential angle of the stress amplification region that is smaller than a circumferential angle of the stress vulnerable region, based on the center of the core.

16. (previously presented)The cylindrical battery according to claim 1,
wherein the winding structure of the electrode assembly is maintained during charging and discharging in a range of 200 cycles to 900 cycles.

17. (previously presented)The cylindrical battery according to claim 1,
wherein the first electrode and the second electrode are a positive electrode and a negative electrode, respectively,

wherein the first electrode includes a plurality of positive electrode winding turns and the second electrode includes a plurality of negative electrode winding turns,

wherein at least a part of the plurality of negative electrode winding turns is provided adjacent to a core side of the electrode assembly, and

wherein at least a part of the plurality of negative electrode winding turns extends in a radial direction.

18. (previously presented) The cylindrical battery according to claim 17,
wherein the separator includes a plurality of winding turns provided at an inner side of the plurality of negative electrode winding turns.

19. (currently amended) The cylindrical battery according to claim 1,
wherein the first electrode and the second electrode are a positive electrode and a negative electrode, respectively,

wherein the core side end of the second electrode extends further in a direction opposite to the winding-axis direction than the core side end of the first electrode, so as to define at least a part of an innermost winding turn, and

wherein the outer circumference side end of the second electrode extends further in the winding-axis direction than the outer circumference side end of the first electrode, so as to define at least a part of an outermost winding turn.

20. (cancelled)

21. (currently amended) The cylindrical battery according to claim ~~20~~1,
wherein in the winding structure, the circumferential angle of the stress vulnerable region is has an angle in a range of 87 degrees or more, and the circumferential angle of the stress amplification region has an angle in a range of 32 degrees or less.

22. (previously presented) The cylindrical battery according to claim 1,

wherein the battery housing includes an open end and a bottom portion opposing thereto and accommodates the electrode assembly in a space between the open end and the bottom portion, and

wherein the battery housing is electrically connected to one of the first electrode or the second electrode, so as to have a first polarity.

23. (previously presented) The cylindrical battery according to claim 22, further comprising:

a sealing body sealing the open end of the battery housing; and

a terminal electrically connected to the other of the first electrode or the second electrode, so as to have a second polarity,

wherein the terminal has a surface exposed to an exterior of the cylindrical battery.

24. (previously presented) The cylindrical battery according to claim 1,

wherein the first electrode includes a first uncoated portion at a long side end and extending along the winding axis direction of the electrode assembly,

wherein the first uncoated portion extends and protrudes outward from the separator through one end of the electrode assembly and is in a bent state extending in a radial direction of the electrode assembly to define a first bending surface region, and

wherein the cylindrical battery further comprises a first current collector coupled to the first bending surface region.

25. (currently amended) The cylindrical battery according to claim 24,

wherein the first uncoated portion includes a plurality of segments along the winding-axis direction of the electrode assembly, and

wherein the plurality of segments are in a bent state extending in the radial direction of the electrode assembly to define the first bending surface region.

26. (previously presented)The cylindrical battery according to claim 24,
wherein the second electrode includes a second uncoated portion at a long side end and extending along the winding axis direction,
wherein the second uncoated portion extends and protrudes outward from the separator through the other end of the electrode assembly and is in a bent state extending toward the core to define a second bending surface region, and
wherein the cylindrical battery further comprises a second current collector coupled to the second bending surface region.

27. (previously presented)The cylindrical battery according to claim 26,
wherein the second uncoated portion includes a plurality of segments along the winding axis direction of the electrode assembly, and
wherein the plurality of segments are in a bent state extending in the radial direction of the electrode assembly to define the second bending surface region.

28. (previously presented)The cylindrical battery according to claim 24,
wherein a coupling region of the first current collector has a radial structure.

29. (previously presented)The cylindrical battery according to claim 26,
wherein the welding a coupling region of the second current collector has a radial structure.

30. (previously presented)The cylindrical battery according to claim 1,
wherein the battery has a ratio of a diameter to a height greater than 0.4.

31. (previously presented)The cylindrical battery according to claim 30,
wherein the cylindrical battery has a form factor of 46110, 4875, 48110, 4880, 4680, or 4695.

32. (previously presented)A battery pack, comprising a plurality of cylindrical batteries according to claim 1.

33. (original) A vehicle, comprising the battery pack according to claim 32.

34. (new) The cylindrical battery according to claim 1, wherein a diameter of the cross-section is greater than 40mm.

REMARKS

The present communication is in response to the Official Action mailed March 25, 2025 (the “Action”). Claims 1-33 were pending in the present application. Of those claims, claims 1, 19, 21, and 25 are amended, claim 20 is cancelled, and new dependent claim 34 is added herein.

Claim 1 is amended to recite “so as to define a hollow core” and “wherein in the winding structure, a circumferential angle of the stress vulnerable region has an angle of 180 degrees or less, and a circumferential angle of the stress amplification region has an angle in a range of 10 degrees to 90 degrees.” Support may be found at least in ¶¶ [0199], [0249], and previously presented claim 20 of the originally filed substitute specification (hereinafter “the application”).

Claims 19, 21, and 25 are amended to address informalities. Newly presented dependent claim 34 recites “wherein a diameter of the cross-section is greater than 40mm.” (See ¶¶ [0032] and [0656] of the application). No new matter is introduced by any of the foregoing amendments.

I. TELEPHONE INTERVIEW

Initially, Applicant and the undersigned would like to thank Examiner Dignan for the courtesies extended during the telephone interview on May 29, 2025. As memorialized in the Interview Summary dated June 2, 2025, during the interview, the Examiner agreed that the proposed amendments, reflected in the current amendment, would overcome the outstanding rejections. Applicant hereby submits a formal response and respectfully requests reconsideration of the rejections in view of the claims as presented herein and the agreement reached during the interview, as discussed in more detail below.

II. DOUBLE PATENTING

The Action provisionally rejected claims 1-33 on the grounds of nonstatutory obviousness-type double patenting rejection in view of claims 1, 3, and 5-20 of copending application No. 18/670,134 (“the ‘134 application”). Without conceding the propriety of such double patenting rejection, a terminal disclaimer is submitted herewith with respect to the ‘134 Application. Accordingly, Applicant respectfully requests that the nonstatutory obviousness-type double patenting rejection with respect to the ‘134 Application be withdrawn.

III. CLAIM REJECTIONS – 35 U.S.C §112

The Action rejected claims 19 and 25 under 35 U.S.C §112(a) as allegedly failing to comply with the written description requirement. Regarding claim 19, the Action alleged that there is no clear support for a “winding axis direction” extension difference. Regarding claim 25, the Action alleged that the use “winding axis direction” regarding the “plurality of segments” is not supported. The Action rejected claims 1-33 under 35 U.S.C §112(b) as allegedly indefinite. More specifically, the Action alleged that the “stress vulnerable region” and “the stress amplification region” are indefinite and are being interpreted by the Examiner as limitations to arbitrary names for sections on the outer circumference (i.e., arcs) of the electrode assembly.

Initially, regarding the §112(a) rejections of claims 19 and 25, Applicant respectfully submits that amended claims 19 and 25 included herein renders these rejections moot, as agreed during the interview. Regarding the §112(b) of claims 1-33, for the purposes of claim interpretation, Applicant does not contest the Examiner’s interpretation under the Broadest Reasonable Interpretation. Accordingly, Applicant respectfully requests that the rejections be withdrawn, as agreed during the interview.

IV. CLAIM REJECTIONS – 35 U.S.C §§ 102 AND 103

The Action made the following rejections:

- Claims 1-3, 13-19, and 22-23 under 35 U.S.C. §102(a)(1) as anticipated by US 2021/0376391 (“*Okuda*”).
- Claims 1-6, 9-15, 17-23, and 30 under 35 U.S.C. §102(a)(1) as anticipated by US 3,298,871 (“*Binder*”).
- Claims 7-8 and 31-33 under 35 U.S.C. §103 over *Binder*.
- Claims 24-29 under 35 U.S.C. §103 over *Okuda* in view of US 2021/0344033 (“*Park*”).

The Action alleged that the cited reference teaches all of the elements of claim 1. Regarding claim 20, the Action alleged that FIG. 1 of *Binder* corresponds to the recited “a circumferential angle of the stress vulnerable region has an angle in a range of 30 degrees to 180 degrees, and a circumferential angle of the stress amplification region has an angle in a range of 10 degrees to 90 degrees.” Applicant respectfully traverses each of the rejections of record.

However, in order to expedite prosecution, claim 1 is amended to recite “so as to define a hollow core” and “wherein in the winding structure, a circumferential angle of the stress vulnerable region has an angle of 180 degrees or less, and a circumferential angle of the stress amplification region has an angle in a range of 10 degrees to 90 degrees.” Support may be found at least in ¶¶ [0199], [0249], and previously presented claim 20 of the originally filed substitute specification (hereinafter “the application”).

In contrast, *Binder* explicitly teaches a mandrel (M) for receiving a portion of the electrode (See Col. 2 Lns. 11-22; Col. 3 Lns. 43-49; Col. 4 Lns. 70-75; FIGS. 1, 4-5 of *Binder*). Furthermore, FIG. 1 of *Binder* cannot be lied upon as teaching or suggesting “the stress vulnerable region has an angle of 180 degrees or less, and a circumferential angle of the stress amplification region has an angle in a range of 10 degrees to 90 degrees” as recited in amended claim 1 because proportions of features in a drawing are not evidence of actual proportions when drawings are not to scale, as in the cited art. (See MPEP 2125 (II); Action PG. 12). For at least these reasons, Applicant respectfully requests withdrawal of the rejections of record, as agreed during the telephone interview.

V. NEW CLAIM 34

Newly presented claim 34 depends from independent claim 1 and recites “wherein a diameter of the cross-section is greater than 40mm.” (See ¶¶ [0032] and [0656] of the application). As generally agreed during the Interview, newly presented claim 34 is believed to further define over the prior art. Favorable consideration is respectfully requested.

VI. CONCLUSION

As it is believed that all of the rejections set forth in the Official Action have been fully met, favorable reconsideration and allowance are earnestly solicited. If, however, for any reason the Examiner does not believe that such action can be taken at this time, it is respectfully requested that the Examiner telephone applicant’s attorney at (908) 654-5000 in order to overcome any additional objections which the Examiner might have.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: June 25, 2025

Respectfully submitted,

Electronic signature:

/Matthew J. Shatynski/

Matthew J. Shatynski

Registration No.: 81,269

LERNER DAVID LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant

AMENDMENT TRANSMITTAL LETTER			Docket No. LGES 3.11F-5465
Application No. 18/848,306-Conf. #7893	Filing Date January 8, 2025	Examiner M. L. Dignan	Art Unit 1723

Applicant(s): LG Energy Solution, Ltd.

Invention: Cylindrical Battery, Battery Pack and Vehicle

TO THE COMMISSIONER FOR PATENTS

Transmitted herewith is an amendment in the above-identified application.
The fee has been calculated and is transmitted as shown below.

CLAIMS AS AMENDED					
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate	
Total Claims	33	- 33 =	0	x	
Independent Claims	1	- 3 =	0	x	
Multiple Dependent Claims (check if applicable) ☐					
Other fee (please specify):					
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00

☒ Large Entity
 ☐ Small Entity

☒ No additional fee is required for this amendment.

☐ Please charge Deposit Account No. _____ in the amount of \$ _____.

☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.

☐ Payment by credit card. Form PTO-2038 is attached.

☒ The Director is hereby authorized to charge and credit Deposit Account No. 12-1095 as described below. A duplicate copy of this sheet is enclosed.

☒ Credit any overpayment.

☒ Charge any additional filing or application processing fees required under 37 C.F.R. § 1.16 and 1.17.

/Matthew J. Shatynski/

Matthew J. Shatynski (Atty/Agent Reg. No.: 81,269)

LERNER DAVID LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Dated: June 25, 2025

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: June 25, 2025

Electronic Signature for Matthew J. Shatynski: / Matthew J. Shatynski /



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Commissioner for Patents
United States Patent and Trademark Office
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APPROVAL LETTER

**APPLICATION #
18/848,306**

**FILING DATE
01/08/2025**

**APPLICANT/PATENT UNDER REEXAMINATION
Yong-Hwan Kwon**

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Electronic terminal disclaimer filed on 06/25/2025



Approved

This patent is subject to a Terminal Disclaimer

Approved / Disapproved by: Electronic Terminal Disclaimer automatically approved



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TERMINAL DISCLAIMER TO OBVIATE A DOUBLE PATENTING REJECTION OVER A PRIOR PATENT

APPLICATION #	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET #
18848306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle



Filing of terminal disclaimer does not obviate requirement for response under 37 CFR 1.111 to outstanding Office Action



This electronic Terminal Disclaimer is not being used for a Joint Research Agreement.

Owner	Percent interest
LG Energy Solution, Ltd.	100%
Total	100%

The owner(s) of percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of any patent granted on pending reference Application Number(s)

Application #	Filing Date
18670134	05/21/2024

as the term of any patent granted on said reference application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending reference application. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and any patent granted on the reference application are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the owner does not disclaim the terminal part of any patent granted on the instant application that would extend to the expiration date of the full statutory term of any patent granted on said reference application, "as the term of any patent granted on said reference application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending reference application," in the event that any such patent granted on the pending reference application: expires for failure to pay a maintenance fee, is held unenforceable, is found invalid by a court of competent jurisdiction, is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321, has all claims canceled by a reexamination certificate, is reissued, or is in any manner terminated prior to the expiration of its full statutory term as shortened by any terminal disclaimer filed prior to its grant.

The owner(s) of percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of prior patent number(s)

Patent #

as the term of said prior patent is presently shortened by any terminal disclaimer. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and the prior patent are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the owner does not disclaim the terminal part of the term of any patent granted on the instant application that would extend to the expiration date of the full statutory term of the prior patent, "as the term of said prior patent is presently shortened by any terminal disclaimer," in the event that said prior patent later:

- expires for failure to pay a maintenance fee;
- is held unenforceable;
- is found invalid by a court of competent jurisdiction;
- is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321;
- has all claims canceled by a reexamination certificate;
- is reissued; or
- is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer.



Terminal disclaimer fee under 37 CFR 1.20(d) included with Electronic Terminal Disclaimer request.

Applicant claims the following entity status:

Regular Undiscounted

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I certify, in accordance with 37 CFR 1.4(d)(4) that I am: An attorney or agent registered to practice before the Patent and Trademark Office who is of record in this application

Signature	Name	Registration #
/Matthew J. Shatynski/	Matthew Shatynski	81269

* Statement under 37 CFR 3.73(b) is required if terminal disclaimer is signed by the assignee (owner). Form PTO/SB/96 may be used for making this certification. See MPEP 324.



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/848,306

RECEIPT DATE / TIME
06/25/2025 03:31:58 PM Z ET

ATTORNEY DOCKET #
LGES 3.11F-5465

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	7893	FILED BY	Matthew Shatynski
PATENT CENTER #	71076037	FILING DATE	01/08/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Yong-Hwan Kwon
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
petition-request.pdf	3	Terminal Disclaimer-Filed (Electronic)	44 KB
grantLetter.pdf	1	Terminal Disclaimer-Electronic-Approved	18 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
petition-request.pdf	CB09E14B5B5499B59DC551EA5EE0AEE04BD1ACBF26A91753 CD0BEF6BFA66B95CEC632F33A415626387FBCB8B05B7C379 2EB8F3A6FDBA0C6DCB924FD5D73B4007

grantLetter.pdf

35A77D88330EBD9E1F4B76ABE69DE11E3BDF1847B75AAC20
34CC5BEA88637DCCB8559399AC305EE65D7C33864C467ACE
75E97C31E0DD6714319BFCC57D7AE0E6

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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ELECTRONIC PAYMENT RECEIPT

APPLICATION # 18/848,306	RECEIPT DATE / TIME 06/25/2025 03:31:58 PM Z ET	ATTORNEY DOCKET # LGES 3.11F-5465
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Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	7893	FILED BY	Matthew Shatynski
PATENT CENTER #	71076037	AUTHORIZED BY	-
CUSTOMER #	86765	FILING DATE	01/08/2025
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Yong-Hwan Kwon

Payment Information

PAYMENT METHOD CARD / 1060	PAYMENT TRANSACTION ID E20256OF32316189	PAYMENT AUTHORIZED BY Matthew Shatynski
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FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1814	STATUTORY DISCLAIMER, INCLUDING TERMINAL DISCLAIMER	183.00	1	183.00
TOTAL AMOUNT:				\$183.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 18/848,306		Filing Date 01/08/2025		☐ To be Mailed		
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO										
APPLICATION AS FILED - PART I										
		(Column 1)		(Column 2)						
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)		
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A				
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =				
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =				
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).								
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL				
APPLICATION AS AMENDED - PART II										
		(Column 1)		(Column 2)		(Column 3)				
AMENDMENT	06/25/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		
								ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	* 33	Minus	** 33	= 0	x \$200 =		0		
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$600 =		0		
	☐ Application Size Fee (37 CFR 1.16(s))									
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0
		(Column 1)		(Column 2)		(Column 3)				
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		
								ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =				
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =				
	☐ Application Size Fee (37 CFR 1.16(s))									
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE		
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/PAMELA V THERATT/		
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".										
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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UNITED STATES DEPARTMENT OF COMMERCE
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NOTICE OF ALLOWANCE AND FEE(S) DUE

86765 7590 07/18/2025
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EXAMINER	
DIGNAN, MICHAEL L	
ART UNIT	PAPER NUMBER
1723	

DATE MAILED: 07/18/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893

TITLE OF INVENTION: Cylindrical Battery, Battery Pack and Vehicle

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	10/20/2025

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893
TITLE OF INVENTION: Cylindrical Battery, Battery Pack and Vehicle				

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	10/20/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
DIGNAN, MICHAEL L	1723	429-094000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893
86765	7590	07/18/2025	EXAMINER	
LG Lerner David LLP 20 Commerce Drive Cranford, NJ 07016			DIGNAN, MICHAEL L	
			ART UNIT	PAPER NUMBER
			1723	
DATE MAILED: 07/18/2025				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 18/848,306	Applicant(s) Kwon et al.	
	Examiner MICHAEL L DIGNAN	Art Unit 1723	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendments filed 6/25/2025.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-19 and 21-34. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____. | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material ____. | 7. ☐ Other ____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date ____. | |

/MICHAEL L DIGNAN/
Examiner, Art Unit 1723

DETAILED ACTION

Notice to Applicant

1. In the amendment dated 6/25/2025, the following has occurred: Claims 1, 21, and 25 have been amended; Claim 20 has been canceled; Claim 34 has been added.
2. Claims 1-19 and 21-34 are pending and are examined herein.
3. The Terminal Disclaimer filed 6/25/2025 is accepted. The Double Patenting rejections are withdrawn.

EXAMINER'S AMENDMENT

4. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given in an interview with Matthew Shatynski on 7/8/2025.

Claim 1 of the application has been amended as follows:

1. (currently amended) A cylindrical battery, comprising:
an electrode assembly in which a first electrode, a second electrode, and a separator interposed therebetween are wound around a winding axis of the electrode assembly, so as to define a hollow core and an outer circumference of the electrode assembly; and
a cylindrical battery housing accommodating the electrode assembly,
wherein a cross-section of the electrode assembly perpendicular to a winding axis direction includes a first fan-shaped region and a second fan-shaped region,

wherein the first fan-shaped region is defined by a first straight line and a second straight line passing from a center of the core through a core side end of the first electrode and a core side end of the second electrode, respectively, and an outer circumference of the first fan-shaped region is defined as a stress vulnerable region, and

wherein the second fan-shaped region is defined by a third straight line and a fourth straight line passing from the center of the core through an outer circumference side end of the first electrode and an outer circumference side end of the second electrode, respectively, and an outer circumference of the second fan-shaped region is defined as a stress amplification region, and

wherein the electrode assembly has a winding structure in which at least the outer circumference side end of the first electrode of the stress amplification region is spaced apart from an inside of the stress vulnerable region along a circumferential direction of the cross-section of the electrode assembly,

wherein in the winding structure, a circumferential angle of the stress vulnerable region has an angle of 30 degrees or more and 180 degrees or less, and a circumferential angle of the stress amplification region has an angle in a range of 1- degrees to 90 degrees.

Reasons for Allowance

5. Claims 1-19 and 21-34 are allowed.

6. The following is an examiner's statement of reasons for allowance: Claim 1 has been amended to require the subject matter of old claim 20 and to also require "a hollow core." Applicant argues in the Remarks filed 6/25/2025 that *Binder* (US Patent No. 3,298,871 to Binder et al.), previously cited in rejection of claim 20, teaches a winding mandrel in the core for receiving the electrodes and winding them. *Okuda*, previously cited for teaching claim 1, does not teach the claimed angles for the stress

Art Unit: 1723

vulnerable and stress amplification regions. Nor does the cited prior art of record appear to teach or render obvious the claimed combination of elements.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

7. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Michael Dignan, whose telephone number is (571) 272-6425. The examiner can normally be reached from Monday to Friday between 10 AM and 6:30 PM. If any attempt to reach the examiner by telephone is unsuccessful, the examiner's supervisor, Tiffany Legette, can be reached at (571)270-7078. Another resource that is available to applicants is the Patent Application Information Retrieval (PAIR). Information regarding the status of an application can be obtained from the (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAX. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, please feel free to contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). Applicants are invited to contact the Office to schedule an in-person interview to discuss and resolve the issues set forth in this Office Action. Although an interview is not required, the Office believes that an interview can be of use to resolve any issues related to a patent application in an efficient and prompt manner.

/MICHAEL L DIGNAN/
Examiner, Art Unit 1723

<i>Examiner-Initiated Interview Summary</i>	Application No. 18/848,306	Applicant(s) Kwon et al.		
	Examiner MICHAEL L DIGNAN	Art Unit 1723	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
MICHAEL L DIGNAN	Primary Examiner	Telephonic
Matthew Shatynski	Attorney	

Date of Interview: 08 July 2025

Issues Discussed:

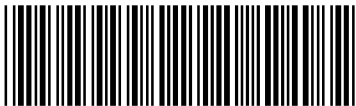
Proposed Amendment(s)

Examiner proposed amendments to make the claims allowable.

/MICHAEL L DIGNAN/ Examiner, Art Unit 1723	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Issue Classification 	Application/Control No. 18/848,306	Applicant(s)/Patent Under Reexamination Kwon et al.
	Examiner MICHAEL L DIGNAN	Art Unit 1723

CPC						
Symbol					Type	Version
H01M	/	50	/	538	F	2021-01-01
H01M	/	50	/	249	I	2021-01-01
H01M	/	50	/	184	I	2021-01-01
H01M	/	50	/	213	I	2021-01-01
H01M	/	10	/	0422	I	2013-01-01
H01M	/	10	/	0431	I	2013-01-01
H01M	/	2220	/	20	A	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	33	
/MICHAEL L DIGNAN/ Examiner, Art Unit 1723	08 July 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	3

Issue Classification 	Application/Control No. 18/848,306	Applicant(s)/Patent Under Reexamination Kwon et al.
	Examiner MICHAEL L DIGNAN	Art Unit 1723

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H01M	/	50	/ 538
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	33	
/MICHAEL L DIGNAN/ Examiner, Art Unit 1723	08 July 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	3

Issue Classification 	Application/Control No. 18/848,306	Applicant(s)/Patent Under Reexamination Kwon et al.
	Examiner MICHAEL L DIGNAN	Art Unit 1723

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	5	19	14	28	24								
2	30	11	6	20	15	29	25								
3	31	12	7	21	16	30	26								
4	32	13	8	22	17	31	27								
5	33	14	9	23	18	32	28								
6	34	15	10	24	19	33	29								
7	2	16	11	25	21										
8	3	17	12	26	22										
9	4	18	13	27	23										

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 33	
/MICHAEL L DIGNAN/ Examiner, Art Unit 1723 (Primary Examiner) _____ (Date) _____		08 July 2025 (Date)	O.G. Print Claim(s) 1 O.G. Print Figure 3

<i>Search Notes</i> 	Application/Control No. 18/848,306	Applicant(s)/Patent Under Reexamination Kwon et al.
	Examiner MICHAEL L DIGNAN	Art Unit 1723

CPC - Searched*		
Symbol	Date	Examiner
H01M 10/0431, 2220/20, 10/0422	03/20/2025	Michael Dignan

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
performed inventor/assignee searches, reviewed IDS and search reports, reviewed global dossier, performed CPC search with text in PE2E	03/20/2025	Michael Dignan
updated search, performed interference search	07/08/2025	Michael Dignan

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H01M	10/0431, 2220/20, 10/0422	07/08/2025	Michael Dignan

/MICHAEL L DIGNAN/ Examiner, Art Unit 1723	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	0	"18848306".apnr.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/17 01:25 PM
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L3	50	(Artificial Intelligence) Similar to: 18/848,306 with 0 CPC Selections and 0 Text Selections Text: (EP-4513664-A1 OR CN-115312877-A OR KR-20230028162-A OR EP-2819204-B1 OR JP-5191993-B2 OR EP-2337120-A2 OR EP-4468495-A1 OR CN-111837255-B OR US-9515309-B2 OR US-20230087166-A1 OR KR-100615153-B1 OR WO-2021006529-A1 OR US-11038218-B2 OR CN-115064757-B	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2025/03/19 02:51 PM

		OR CN-118975031-A OR US-20100143787-A1 OR DE-102018204165-A1 OR CN-222088800-U OR WO-2008056908-A1 OR KR-20240169982-A OR US-20240356126-A1 OR CN-116420273-A OR CN-118117250-A OR CN-119096403-A OR US-20250070218-A1 OR EP-4394996-A1 OR CN-105374956-A OR US-8114536-B2 OR KR-20210006712-A OR CN-102005602-A OR US-20230079811-A1 OR US-9070929-B2 OR EP-2772960-A1 OR CN-222126934-U OR CN-118610669-A OR WO-2024021025-A1 OR EP-4297144-A1 OR CN-219959089-U OR US-20190131594-A1 OR KR-20150055775-A OR KR-100509747-B1 OR CN-116783723-A OR EP-2772960-B1 OR US-20140050960-A1 OR KR-20240170003-A OR KR-20190046404-A OR WO-2018071825-A3 OR KR-101653305-B1 OR EP-4443632-A1 OR CN-114069167-B).did.					
L4	9	("20050031950" OR "20160218368" OR "20070196730" OR "20160351966" OR "20170271640" OR "20210344033" OR "20210376391" OR "6677082" OR "6680143").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/19 02:54 PM
L5	267177	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL,	OR	ON	ON	2025/03/20 10:40 AM

L6	13474	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc. AND jelly ADJ roll	PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/20 10:40 AM
L7	2522	(H01M10/0431 OR H01M2220/20 OR H01M10/0422).cpc. AND jelly ADJ roll	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 10:40 AM
L8	1	(17/282094).APP.	(USPAT; USOCR)	OR	ON	ON	2025/03/20 10:51 AM
L9	8	("20030186095" "20130266834" "20140045036" "20140141303" "20150353396" "20160218368" "20180040881" "20210119263").pn. OR ("12155041").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 10:51 AM
L10	1020	L2 AND (roll\$3 wind\$3 wound) SAME (arc fan)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:16 AM
L11	860	L2 AND (roll\$3 wind\$3 wound) SAME (arc fan) AND (end)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:16 AM
L12	869	L5 AND (roll\$3 wind\$3 wound) SAME (arc fan) AND (end)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:45 AM
L13	3	L5 AND (roll\$3 wind\$3 wound) SAME (arc fan) AND (end) AND ("500" ADJ cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:45 AM
L14	4	L5 AND ("500" ADJ	(US-PGPUB; USPAT;	OR	ON	ON	2025/03/20

		cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	USOCR)				11:49 AM
L15	17	L5 AND (retention SAME cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:50 AM
L16	51	h01m AND (retention SAME cycle) AND ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:52 AM
L17	166	h01m AND (cell form) WITH ("46110" "4875" "48110" "4880" "4680" "4695")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:54 AM
L18	3356	L5 AND (cycles) SAME ("200" "300" "400" "500" "600" "700" "800" "900")	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:56 AM
L19	1299	L5 AND (cycles) SAME ("200" "300" "400" "500" "600" "700" "800" "900") AND cylindrical AND (wound wind\$3 roll\$3)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 11:57 AM
L20	3	"5965290".pn. "6391488".pn. "20110014509".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L21	0	(12/835020).APP.	(USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L22	7	("2011/0014509").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L23	21	("4963445" "5965290").pn. OR ("6391488").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L24	18	("3298871" "4707421" "5354629" "5458993" "5508122").pn. OR ("5965290").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L25	42	L22 L23 L24	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:04 PM
L26	2522	L5 AND (jelly ADJ roll)	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:05 PM
L27	12284	L5 AND (jelly ADJ roll)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA,	OR	ON	ON	2025/03/20 12:05 PM

			EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO))				
L28	8780	L5 AND (jelly ADJ roll) AND cylindrical	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO))	OR	ON	ON	2025/03/20 12:05 PM
L29	1684	L5 AND (jelly ADJ roll) AND cylindrical	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:05 PM
L30	396	L5 AND (jelly ADJ roll) AND cylindrical AND core	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/03/20 12:05 PM
L31	2	"18670134"	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/07/02 11:35 AM
L32	0	(18/670134).APP.	(USPAT; USOCR)	OR	ON	ON	2025/07/02 11:35 AM
L33	430	L5 AND (jelly ADJ roll) AND cylindrical AND core	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2025/07/02 12:07 PM

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	33686	(H01M10/0431 OR	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20

		H01M2220/20 OR H01M10/0422).cpc.					10:52 AM
N2	2	N1 AND cylindrical.clm. AND (fan ADJ shaped).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:52 AM
N3	0	N1 AND cylindrical.clm. AND (arc).clm. AND stress\$3.clm. AND (wound roll).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:53 AM
N4	0	N1 AND cylindrical.clm. AND (arc).clm. AND stress\$3.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/20 10:53 AM
N5	3	N1 AND cylindrical.clm. AND (fan ADJ shaped).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/07/02 12:06 PM
N6	3	N1 AND cylindrical.clm. AND (fan ADJ shaped).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/07/02 12:06 PM
N7	1	N1 AND cylindrical.clm. AND (arc).clm. AND stress\$3.clm.	(US-PGPUB; USPAT)	OR	ON	ON	2025/07/02 12:07 PM

TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Fri Jul 18 05:00:55 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

LG

Lerner David LLP

20 Commerce Drive

Cranford, NEW JERSEY, 07016

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Application	Document	Mailroom Date	Attorney Docket No.
18848306	NOA	07/18/2025	LGES 3.11F-5465
18848306	EXIN	07/18/2025	LGES 3.11F-5465

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If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

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Sincerely,

The Patent Center Team



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/848,306

RECEIPT DATE / TIME
08/11/2025 01:01:34 PM Z ET

ATTORNEY DOCKET #
LGES 3.11F-5465

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	7893	FILED BY	Sharon McGee
PATENT CENTER #	71795390	FILING DATE	01/08/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Yong-Hwan Kwon
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Kevin Kocun

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
LGES5310 - Fee Transmittal -- Part B PTOL-85.pdf	1	Issue Fee Payment (PTO-85B)	39 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
LGES5310 - Fee Transmittal -- Part B PTOL-85.pdf	A9DE86FF016C1A873CD9DAE890633E21E889EC9FE044E1B2 C99D0958ADAD3A8A072CF04A3908A3003387C0EED847EF03 0178DBEDEB29781C330CE111A917E1BD

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B-FEE(S) TRANSMITTAL

Complete and send this form, together with the applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

By fax, send to: (571) 273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

Matthew J. Shatynski

(Typed or printed name)

/ Matthew J. Shatynski /

(Signature)

August 11, 2025

(Date)

APPLICATION NO	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO	CONFIRMATION NO
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893

TITLE OF INVENTION: Cylindrical Battery, Battery Pack and Vehicle

APPLN: TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1,290.00			\$1,290.00	10/18/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
M. L. Dignan	1723	429-094000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363)

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1. Lerner David LLP

2.

3.

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE or COUNTRY)

LG Energy Solution, Ltd.

Seoul, Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent):

☐ Individual ☒ Corporation or other private group entity ☐ Government4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment (Please first reapply any previously paid fee shown above):

- ☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)
- ☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 12-1095

5. Change of Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29. NOTE: Absent a valid Certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.
- ☐ Applicant asserting small entity status. See 37 CFR 1.27. NOTE: If the application was previously under micro entity status, checking this box will be taken as a notification of loss of entitlement to micro entity status.
- ☐ Applicant changing to regular undiscounted fee status. NOTE: Checking this box will be taken as a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature

/ Matthew J. Shatynski /

Date August 11, 2025

Typed or printed name

Matthew J. Shatynski

Registration No.

81,269



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/848,306	08/11/2025 01:01:34 PM Z ET	LGES 3.11F-5465

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	7893	FILED BY	Sharon McGee
PATENT CENTER #	71795390	AUTHORIZED BY	Kevin Kocun
CUSTOMER #	86765	FILING DATE	01/08/2025
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Yong-Hwan Kwon

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 1011	E20258AD02029602	Sharon McGee

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

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New Applications Under 35 U.S.C. 111

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National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	09/09/2025	12412965	LGES 3.11F-5465	7893

86765 7590 08/27/2025

LG

Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Yong-Hwan Kwon, Daejeon, KOREA, REPUBLIC OF;
Dong-Wook Kim, Daejeon, KOREA, REPUBLIC OF;
Ji-Young Park, Daejeon, KOREA, REPUBLIC OF;
Ju-Nam Son, Daejeon, KOREA, REPUBLIC OF;
Seo-Hyeon Chae, Daejeon, KOREA, REPUBLIC OF;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF;

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TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Thu Aug 28 05:02:50 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

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Application	Document	Mailroom Date	Attorney Docket No.
18848306	ISSUE.NTF	08/27/2025	LGES 3.11F-5465

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If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893
86765	7590	09/09/2025		
LG Lerner David LLP 20 Commerce Drive Cranford, NJ 07016			EXAMINER DIGNAN, MICHAEL L	
			ART UNIT	PAPER NUMBER
			1723	
			NOTIFICATION DATE	DELIVERY MODE
			09/09/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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APPLICATION NO.	ISSUE DATE	PATENT NO.
18/848,306	09-Sep-2025	12412965

LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EGRANT NOTIFICATION

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TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - eGrant Notification for Customer Number 86765

Tue Sep 09 06:00:52 EDT 2025

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Application	Patent	Document	Mailroom Date	Attorney Docket No.
18848306	12412965	EGRANT.NTF	09/09/2025	LGES 3.11F-5465

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Dated: January 26, 2026
Electronic Signature for Matthew J. Shatynski: /Matthew J. Shatynski/

Docket No. LGES 3.11F-5465
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Letters Patent of:
Kwon et al.

Patent No.: 12,412,965

Issued: September 9, 2025

For: Cylindrical Battery, Battery Pack and Vehicle

Certificate of Correction Branch
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REQUEST FOR CERTIFICATE OF CORRECTION PURSUANT TO 37 C.F.R. § 1.323

Dear Commissioner

Enclosed herewith is a Certificate of Correction with respect to the above-identified U.S. Patent. The corrections indicated should be made as the typographical/clerical error was made by the Patent Office in the Examiner's Amendment in the Notice of Allowance mailed July 18, 2025. There is no requirement for payment of a fee since the errors were made by the Patent Office. Applicant believes no fee is due. However, if a fee is due, please charge our Deposit Account No. 12-1095.

We look forward to early return of the Certificate of Correction duly certified.

Dated: January 26, 2026

Respectfully submitted,
Electronic signature:
/Matthew J. Shatynski /
Matthew J. Shatynski
Registration No.: 81,269
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

**UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION**

Page 1 of 1

PATENT NO. : 12,412,965
APPLICATION NO. : 18/848,306
ISSUE DATE : September 9, 2025
INVENTOR(S) : Yong-Hwan Kwon, Dong-Wook Kim, Ji Young Park, Ju-Nam Son, and Seo-Hyeon Chae

It is certified that an error appears or errors appear in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Column 105, line 16, "1-degrees" should read --10 degrees--

MAILING ADDRESS OF SENDER (Please do not use Customer Number below):

Matthew J. Shatynski
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016



UNITED STATES
PATENT AND TRADEMARK OFFICE

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Alexandria, VA 22313 - 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/848,306

RECEIPT DATE / TIME
01/26/2026 02:01:21 PM Z ET

ATTORNEY DOCKET #
LGES 3.11F-5465

Title of Invention

Cylindrical Battery, Battery Pack and Vehicle

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	12412965
CONFIRMATION #	7893	FILED BY	Grace Liu
PATENT CENTER #	74133737	FILING DATE	01/08/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Yong-Hwan Kwon
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	James Vezeris

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
LGES5310 - Certificate Of Correction - PTO Error LD-449.pdf	1	Request for Certificate of Correction	23 KB
LGES5310 - Certificate of Correction PTO SB-44.pdf	1	Request for Certificate of Correction	14 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
LGES5310 - Certificate Of Correction - PTO Error LD-	A4C12E6BB9527B92AB54264176228A3EB85FF5E58946E8D5742B0088494658027D21071993E10DD806DC880517CB835EFC4

449.pdf

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Correction PTO SB-44.pdfC470F65A70908E6B127F361EB33F5BE98D4218E01B474DB4E
FB39323B3FCCF04D64500D6369D40EDFDCBAFBC30B8E66A
C5B042814A4A76AFB90482AA40B563FB

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New Applications Under 35 U.S.C. 111

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UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 12,412,965 B2
APPLICATION NO. : 18/848306
DATED : September 9, 2025
INVENTOR(S) : Yong-Hwan Kwon et al.

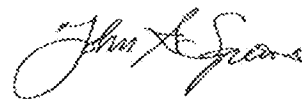
Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Column 105, Line 16, Claim 1, "1-degrees" should read --10 degrees--

Signed and Sealed this
Twenty-fourth Day of February, 2026

A handwritten signature in cursive script, appearing to read "John A. Squires".

John A. Squires
Director of the United States Patent and Trademark Office



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/848,306	01/08/2025	Yong-Hwan Kwon	LGES 3.11F-5465	7893
86765	7590	02/24/2026	EXAMINER	
LG			DIGNAN, MICHAEL L	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1723	
			NOTIFICATION DATE	DELIVERY MODE
			02/24/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

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APPLICATION NO.	ISSUE DATE	PATENT NO.
18/848,306	24-Feb-2026	12412965

LG
Lerner David LLP
Cranford, NJ 07016

ELECTRONIC CERTIFICATE OF CORRECTION NOTIFICATION

Your electronic certificate of correction is now available, and can be accessed via Patent Center at <https://patentcenter.uspto.gov>.

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TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 86765

Tue Feb 24 05:08:36 EST 2026

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Application	Document	Mailroom Date	Attorney Docket No
18848306	COCOUT	02/24/2026	LGES 3.11F-5465

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Sincerely,

The Patent Center Team

TO: eOfficeAction@lernerdavid.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - eGrant Notification for Customer Number 86765

Tue Feb 24 06:06:09 EST 2026

Dear Patent Center Customer:

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Application	Patent	Document	Mailroom Date	Attorney Docket No.
18848306	12412965	ECOF.C.NTF	02/24/2026	LGES 3.11F-5465

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